

Research paper

Boost DC-DC converter with regenerative snubber

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ABSTRACT

To accommodate the wide range of input voltages supplied by redundant batteries and ensure an adequate hold-up time for communication systems during utility power failures, power supplies used in 5 G base stations typically include a front-end boost converter as the first stage and a second-stage isolated converter. In this paper, a new topology for a regenerative snubber in boost converters is proposed. Compared to traditional active or passive snubber topologies, the proposed snubber offers the advantages of having fewer components and occupying less space. Moreover, it does not require additional active devices and can be driven by widely-used commercial PWM control integrated circuits, achieving a higher level of cost-effectiveness and offering increased industrial applications. The final prototype of a power supply for a networking switch hub (as used in 5 G base stations) is fabricated to demonstrate the feasibility of the proposed method.

1. Introduction

Since 5 G communication operates within a high-frequency band, it is easily blocked by buildings. In urban areas, the average distance between 5 G cell stations ranges from 100 to 300 m, meaning a large number of cell stations are required. Accordingly, given that these stations are often installed on traffic lights, utility poles, or street lights, the base station equipment needs to be compact and highly efficient.

Power supplies used in 5 G base stations typically consist of a front-end boost converter (the first stage) and an isolated converter (the second stage). The boost converter can meet the hold-up time requirement for communication systems during utility power failures. The isolated second-stage converter can meet the high-potential test and leakage current requirements.

Fig. 1(a) displays a traditional boost converter. Here, the main switch employs hard switching that generates significant switching losses, which can be decreased by using soft switching technology. Many previous topologies of boost converters with soft switching have been proposed. However, they still face challenges in terms of industrial application. Table 1 presents a comparison of the proposed research and references. The methods in (Dong et al., 2002; Bodur et al., 2013; Irving and Jovanovic, 2002; Li et al., 2010; Shamsi et al., 2021; Sahin et al., 2018; Zhan et al., 2014; Kim et al., 2015) require additional high-rated active or passive components, which are typically expensive and bulky. In addition, the circuits used in (Kim et al., 2015; Mohammadi et al., 2015; Cho et al., 2010) contain an excessive number of components,

rendering it difficult to fit them into a compact space. The methods proposed in (Shamsi et al., 2021) and (Esteki et al., 2017) have fewer components and incorporate soft switching for turning-on and turning-off functions. However, their conversion efficiency decreases significantly under light load conditions or when the input voltage decreases.

The previous solution presented in Fig. 1(b) (Yau and Hung, 2023) is both simple and cost-effective. However, it only provides a soft-switching function on the main switch during turn-off, not during turn-on. In comparison, the proposed snubber only requires one additional small inductor to enable a turn-on soft-switching function.

Fig. 2 displays the proposed topology. The original boost converter includes switch Q_1 , which comprises output capacitors C_{oss1} , input capacitor C_{in} , output capacitor C_o , output diode D_1 , and main inductor L_1 . The snubber consists of diodes D_3 and D_5 , inductor L_s , inductor L_3 , and capacitor C_1 , as displayed in Fig. 2. In this paper, the proposed circuit is presented in detail, complete with the relevant equations, a mode analysis, and the operation states. Finally, a prototype circuit for powering networking equipment is fabricated and measured to verify the effectiveness of the proposed method.

2. Proposed circuit configuration

First, it is necessary to define the symbols and variables, which are displayed in Table 2.

The proposed circuit has four operation modes (Modes I to IV), as

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depicted in Fig. 3. These four modes change depending on the current operating conditions, which are determined by variations in T_{on} and inductor currents resulting from different input voltages and load currents. Fig. 3 illustrates the switching mechanism of the operating modes through a state machine. When the specified conditions are met, a transition will occur from the current operating mode to another mode. However, the controller of the converter will not intervene or detect the mode change.

Before analyzing the four operating modes, the first step is to assume that the converter operates in continuous conduction mode (CCM). For convenience, when explaining the operation states, all of the components are assumed as ideal models (i.e., $i_{Ls}(t_0)$, $i_{L3}(t_0)$, and $i_{C1}(t_0)$ are assumed to be 0 A at t_0). This means that all energy stored in L_s and L_3 is reset to zero before each switching period starts.

Since the inductance of L_3 is much smaller than that of L_s or L_1 , when L_3 is connected in series with L_s or L_1 , the voltage across L_3 is divided and becomes extremely low. During Q_1 is turn-off, it is assumed that the capacitance of C_1 is much higher than C_{oss1} of Q_1 , so the effect of C_{oss1} can be ignored when L_s resonates with C_1 and C_{oss1} .

$$\omega_0 = \frac{1}{\sqrt{C_1 \cdot L_s}} \quad (1)$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 \cdot L_s}} \quad (2)$$

$$T_0 = 2\pi\sqrt{C_1 \cdot L_s} \quad (3)$$

$$Z_0 = \sqrt{\frac{L_s}{C_1}} + R_{Ls} + R_{C1} + L_{C1} \quad (4)$$

$$\omega_1 = \frac{1}{\sqrt{(C_1 + C_{oss1})(L_1 + L_3)}} \quad (5)$$

$$Z_1 = \sqrt{\frac{L_1 + L_3}{C_1 + C_{oss1}}} + R_{L1} + R_{C1} + L_{C1} \quad (6)$$

$$\omega_2 = \frac{1}{\sqrt{(C_1 + C_{oss1}) \cdot L_3}} \quad (7)$$

$$T_2 = 2\pi\sqrt{(C_1 + C_{oss1}) \cdot L_3} \quad (8)$$

$$Z_2 = \sqrt{\frac{L_3}{C_1 + C_{oss1}}} + R_{L3} + R_{C1} + L_{C1} \quad (9)$$

$$i_{L1}(t_0) = i_{D1}(t_0) = \frac{P_{in}}{V_{in}} - \frac{V_{in} \cdot T_{on}}{2 \cdot L_1} \quad (10)$$

$$i_{Ls}(t_0) = 0 \quad (11)$$

$$i_{L3}(t_0) = 0 \quad (12)$$

$$i_{C1}(t_0) = 0 \quad (13)$$

$$i_{D5}(t_0) = 0 \quad (14)$$

Finally, it is also assumed that the inductance of L_3 is significantly smaller than the inductances of L_1 and L_s . Therefore, v_{L3} can be treated as zero to simplify the derivation when L_3 is cascaded with L_1 or L_s .

2.1. Operation mode I

The simulation waveforms of Mode I are displayed in Fig. 4. First, there are some initial conditions, as displayed in (12)–(15).

$$v_{C1}(t_0) = -\sqrt{\frac{L_3 \cdot (i_{L1-pk})^2}{C_1}} \quad (15)$$

1) State 1 (t_0-t_1): As displayed in Fig. 5(a), Q_1 is turned on. The initial value of i_{D1} is expressed in (10). In this state, i_{L3} increases from zero but is still less than i_{L1} , indicating that there is a partial current flowing through D_1 (termed i_{D1}), as expressed in (22). L_1 is demagnetized to decrease i_{L1} . V_{in} charges the resonant tank consisting of L_s and C_1 . i_{Ls} and v_{C1} can be expressed as (17) and (20), respectively. Simultaneously, i_{L3} and i_{ds1} start to increase from zero, as displayed in (18) and (21), to decrease the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . As a result, the turn-on switching loss is reduced. The rise time of i_{ds1} is calculated as shown in (23). When i_{D1} reduces to zero, the converter transitions to State 2.

$$i_{L1}(t) = i_{L1}(t_0) + \left(\frac{V_{in} - V_o}{L_1}\right)(t - t_0) \quad (16)$$

$$i_{Ls}(t) = i_{C1}(t) = i_{Ls}(t_0) + \left(\frac{V_{in} - v_{C1}(t_0)}{Z_0}\right) \cdot \sin \omega_0(t - t_0) \quad (17)$$

$$i_{L3}(t) = i_{L3}(t_0) + \left(\frac{V_o - 0}{L_3}\right) \cdot (t - t_0) = \left(\frac{V_o - 0}{L_3}\right) \cdot (t - t_0) \quad (18)$$

$$i_{L3}(t_1) = i_{L1}(t_1) \quad (19)$$

$$v_{C1}(t_1) = V_{in} - (V_{in} - v_{C1}(t_0)) \cdot \cos \omega_0(t_1 - t_0) \quad (20)$$

$$i_{ds1}(t) = i_{Ls}(t) + i_{L3}(t) \quad (21)$$

$$i_{D1}(t) = i_{L1}(t) - i_{L3}(t) \quad (22)$$

$$\Delta t_1 = (t_1 - t_0) = \frac{L_3 \cdot i_{L1}(t_0)}{V_o} \quad (23)$$

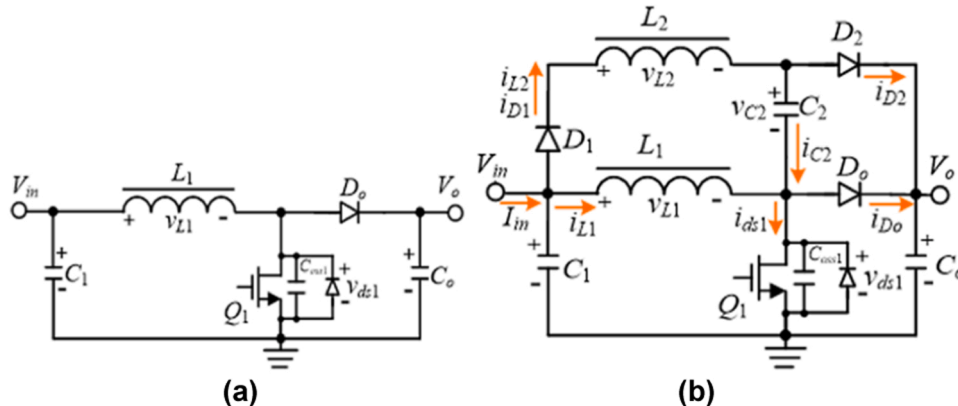
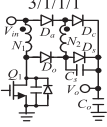
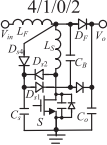
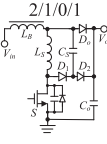
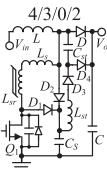


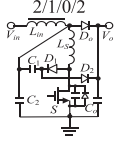
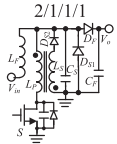
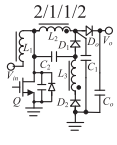
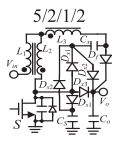
Fig. 1. Boost converters: (a) Conventional hard switching topology, and (b) boost converter with turn-off snubber in (Yau and Hung, 2023).

Table 1
Comparison of proposed topology and references.

Ref. No.	Extra diode/ Extra inductor/ Aux winding/ Extra capacitor	Peak voltage Stress on main switch	Peak current stress on main switch	Voltage stress on diode	Peak current Stress on diode	Condition to reach soft turning-on	Condition to reach soft turning- off	System specification and efficiency	(✓) Advantages and (✗) disadvantages
(Dong et al., 2002)	3/1/1/1 	V_o	$i_{Lm} + \frac{nV_{in}}{\sqrt{\frac{L_k}{C_s}}}$	V_o	$\frac{i_{Lm}}{n}$	$t_f = \frac{V_o + (n-1)V_{in}}{L_k \Delta i_{Lk}}$	$t_r = \pi \sqrt{L_k \cdot C_s}$	AC-DC PFC Input: 90–265 Vac Output: 400 Vdc, 1 kW F_{sw} : 100 kHz η : 96.4 % at 200 Vac, full load η : 91.5 % at 100 Vac, full load	✓ Soft switching on both turn-on and turn-off ✗ Three snubber components located on main power paths ✗ One extra external inductor with high current rating, resulting in increased cost and volume
(Bodur et al., 2013)	4/1/0/2 	V_o	$\sqrt{i_{rr}^2 + \frac{V_o^2 C_e}{L_s}} + I_{in}$	$\sqrt{\frac{L_s I_{rr}^2 + C_s V_o^2}{C_B}} + V_o$	I_{in}	$t_f = \frac{L_s \cdot I_{inpeak}}{V_o}$	$t_r = \frac{C_s \cdot V_o}{I_{inpeak}}$	AC-DC PFC Input: 220 Vac Output: 400 Vdc, 1 kW F_{sw} : 50 kHz η : 97.5 % at 220 Vac, full load	✓ Soft switching on both turn-on and turn-off ✓ One snubber component located on main power paths ✗ One extra inductor with high current rating, resulting in increased cost and volume
(Irving and Jovanovic, 2002)	2/1/0/1 	V_o	$I_{in} + I_{rr}$	V_o	I_{in}	$t_f = \frac{L_s (I_{in} + I_{rr})}{V_o}$	$t_r = \pi \sqrt{L_s \cdot (C_s + C_{oss})}$	AC-DC PFC Input: 90–264 Vac Output: 380 Vdc, 500 W F_{sw} : 80 kHz η : 93.8 % at 200 Vac, full load	✓ Soft switching on both turn-on and turn-off ✓ One snubber component located on main power paths ✗ One extra inductor with high current rating, resulting in increased cost and volume
(Li et al., 2010)	4/3/0/2 	V_o	$V_o \sqrt{\frac{C_s}{L_s + L_{st}}} \sqrt{1 - \frac{C_s}{C_{st}}} + I_{in}$	—	—	$t_f = \frac{2\Delta Q_{rr}}{I_{rr}} + \frac{(L_s + L_{sr})I_{sat}}{V_{out}} + \frac{L_s(I_{in} - I_{sat})}{V_{out}}$	$t_r = \frac{C_s V_{out}}{I_{in}} \left(1 + \frac{1}{\sqrt{x}}\right)$	AC-DC PFC Input: 90–264 Vac Output: 380 Vdc, 750 W F_{sw} : 100 kHz η : 94.5 % at 115 Vac, full load η : 97.6 % at 230 Vac, full load	✓ Soft switching on both turn-on and turn-off ✗ Two snubber components located on main power paths ✗ Two extra inductors with high current rating, resulting in increased cost and volume

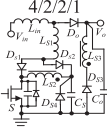
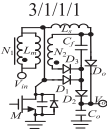
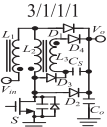
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Table 1 (continued)

Ref. No.	Extra diode/ Extra inductor/ Aux winding/ Extra capacitor	Peak voltage Stress on main switch	Peak current stress on main switch	Voltage stress on diode	Peak current Stress on diode	Condition to reach soft turning-on	Condition to reach soft turning- off	System specification and efficiency	(✓) Advantages and (✗) disadvantages
(Shamsi et al., 2021)	2/1/0/2 	V_o	$V_{in} \cdot \sqrt{\frac{C_2}{L_s}}$ $+I_{in}$	V_o	—	$t_{ri} = \frac{I_{SW,SW} \cdot L_s}{V_{SW,ON}}$	$t_f = \frac{2V_{SW,OFF} \cdot C_2}{I_{SW,OFF}}$	DC-DC boost Input: 28–63 Vdc Output: 96 Vdc, 220 W F_{sw} : 100 kHz η : 96 % at 48 Vdc, full load η : 88 % at 28 Vdc, full load	in increased cost and volume ✓ Soft switching on both turn-on and turn-off ✓ One extra inductor needed ✗ High- inductance and high-cost resonant inductor required ✗ Lower efficiency when low input voltage or light load
(Sahin et al., 2018)	2/1/1/1 	$V_o + \frac{V_o}{n}$	$\frac{\sqrt{V_o^2 + Z^2 I_r^2}}{Z}$ $+I_{in}$ $Z = \sqrt{\frac{L_p}{C_s}}$	$V_o + nV_o$	$I_i + I_{Lsmax}$	$t_f = \frac{L_p \cdot I_i}{V_o}$	$t_r = \frac{C_s \cdot V_o}{I_i + I_{Ls \max}}$	DC-DC boost Input: 50 Vdc Output: 100 Vdc, 200 W F_{sw} : 100 kHz η : 96 %, full load	✓ Soft switching on both turn-on and turn-off ✗ One extra coupled inductor needed ✗ High voltage stresses on diodes
(Zhan et al., 2014)	2/1/1/2 	V_o $+ A \sin(\omega_r t_3(t-t_4))$	$I_{L1} + I_{L2} + I_{C2}$	—	—	$t_f = \frac{I_{L1} L_2}{V_o}$ $+ \sqrt{L_2 C_1} \cdot \arccos\left(\frac{1}{N_1^2}\right)$ $+ \sqrt{L_2 C_1} (1 + N_1^3)$ $\cdot \arctan\left(\sqrt{\frac{L_2 \omega_r^2 C_1 (1 + N_1^6)}{N_1^4 - 1}}\right)$	$t_r = \frac{I_{L30} L_3}{V_o}$ $+ \sqrt{L_3 C_1 + L_2 C_2} \arcsin\left(\frac{V_o}{AN_1}\right)$	DC-DC boost Input: 45–55 V dc Output: 100 Vdc, 400 W F_{sw} : 30 kHz η : 93.6 % at 45 Vdc, full load	✓ Soft switching on both turn-on and turn-off ✗ Snubber components located on main power paths ✗ One extra coupled inductor needed ✗ High voltage and current stresses on main switch ✗ High voltage stresses on diodes ✗ Lower efficiency when low input voltage or light load
(Kim et al., 2015)	5/2/1/2 	V_o	$I_S = I_{L1}(t_1)$ $- \left(1 + \frac{L_2}{L_1}\right) I_{L2}(t_2)$	$V_o + \frac{N_3}{N_1} V_{in}$	i_{L1}	$t_f = \frac{L_3 I_{in}}{V_{out}}$ $\tan^{-1} \left[\frac{C_s \omega_1 V_{out}}{I_{r1} (1 - 2L_2 C_x \omega_1^2)} \right]$ $+ \frac{1}{\omega_1}$	$t_r = \frac{V_{out}}{\sin^{-1} \left(\frac{V_{out}}{I_{r3} Z_3 + \omega_3 V_{out} (1 - D)} \right) \omega_3}$	AC-DC PFC Input: 220 Vac Output: 380 Vdc, 3.3 kW F_{sw} : 50 kHz η : 97.4 % at 220 Vac, full load	✓ Soft switching on both turn-on and turn-off ✗ Snubber components located on main power paths ✗ Two extra inductor windings

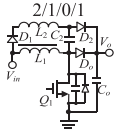
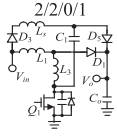
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Table 1 (continued)

Ref. No.	Extra diode/ Extra inductor/ Aux winding/ Extra capacitor	Peak voltage Stress on main switch	Peak current stress on main switch	Voltage stress on diode	Peak current Stress on diode	Condition to reach soft turning-on	Condition to reach soft turning- off	System specification and efficiency	(✓) Advantages and (✖) disadvantages
(Mohammadi et al., 2015)	4/2/2/1 	$V_o \left(1 + \sqrt{\frac{L_{s2}}{L_{s3}}} \right) + I_o \cdot \sqrt{\frac{L_{s1}}{C_s}}$	$V_o \sqrt{\frac{L_{s1}}{C_s}} + I_o$ $+ I_o \sqrt{\frac{L_{s1}}{L_{s2}}}$	V_o	$i_{sw} = \frac{i_{in}}{D} \left(1 + \sqrt{\frac{L_{s1}}{L_{s2}}} \right) - \frac{\Delta I_L}{2} \left(1 - \sqrt{\frac{L_{s1}}{L_{s2}}} \right) + V_o \sqrt{\frac{C_s}{L_{s2}}}$	$t_f = \frac{C_s \cdot 2V_{sw}}{i_{sw}}$	$t_r = \frac{L_{s1} \cdot i_{sw}}{V_{sw}}$	DC-DC boost Input: 50 V dc Output: 100 Vdc, 200 W F_{sw} : 100 kHz η : 95.9 % at 48 Vdc, full load	✖ More cost and volume from additional components ✓ Soft switching on both turn-on and turn-off ✖ Snubber components located on main power paths ✖ One extra inductor and one coupled inductor ✖ More cost and volume from higher rating components
(Cho et al., 2010)	3/1/1/1 	V_o	$\frac{V_{in}}{n(Z_{cf} + Z_{lk})} + I_{in}$	V_o	i_{Lm}	NA	$t_r = \frac{C_{oss} \cdot V_{ds1}}{I_{Lm}}$	AC-DC PFC Input: 90–264 Vac Output: 400 Vdc, 700 W F_{sw} : 65 kHz η : 93.7 % at 115 Vac, full load η : 96.94 % at 220 Vac, full load	✖ Snubber components located on main power paths ✖ Extra inductor and coupled inductor ✖ More cost and volume from higher rated components ✖ Soft switching only on turn-off edge
(Esteki et al., 2017)	3/1/1/1 	V_o	i_1	$V_o + V_{in} \frac{n_2}{n_1}$	$I_1 + 1 - \frac{n_2}{n_1}$	$t_{ri} = \frac{I_{sw} \cdot L_{lk2}}{V_{sw}}$	$t_f = \frac{V_{sw} \cdot C_s}{I_{sw}}$	DC-DC boost Input: 48 V dc Output: 100 Vdc, 200 W F_{sw} : 100 kHz η : 95.8 % at 48 Vdc, full load	✓ Soft switching on both turn-on and turn-off ✓ One multi-winding inductor needed ✓ Less cost and volume from less components

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Table 1 (continued)

Ref. No.	Extra diode/ Extra inductor/ Aux winding/ Extra capacitor	Peak voltage Stress on main switch	Peak current stress on main switch	Voltage stress on diode	Peak current Stress on diode	Condition to reach soft turning-on	Condition to reach soft turning- off	System specification and efficiency	(✓) Advantages and (✗) disadvantages
(Yau and Hung, 2023)	2/1/0/1 	V_o	i_{L1}	V_o	i_{L1}	NA	$t_r = \frac{C_2 \cdot V_{C2(\text{Model})}}{i_{L1-pk}}$ In the region of Mode I $t_r = \frac{C_2 \cdot V_{C2(\text{Model})}}{i_{L1-pk}}$ In the region of Mode II	DC-DC boost Input: 20–60 Vdc Output: 48.5Vdc, 340 W F_{sw} : 100 kHz η : 94.72 % at 24 Vdc, full load η : 96.62 % at 36 Vdc, full load	✗ Lower efficiency when low input voltage or light load ✓ Snubber components not located on main power paths ✓ One small extra inductor needed ✓ Less cost and volume from low- rating components ✓ High efficiency when low input voltage or light load ✗ Soft switching only on turn-off edge
Proposed circuit	2/2/0/1 	$\sqrt{\frac{L_3(I_{L1-pk})^2}{C_1}} + V_o$	$i_{L1} + i_{L3}$	V_o	i_{L1}	$t_{ri} = \frac{I_{L1-pk} \cdot L_3}{V_o}$	$t_{rv} = \frac{C_1 \cdot V_{C2(\text{Model})}}{i_{L1-pk}}$ In the region $+\pi\sqrt{L_3 \cdot C_1}$ In the region of Mode I $t_{rv} = \frac{C_1 \cdot V_{C2(\text{Model})}}{i_{L1-pk}}$ In the region $+\pi\sqrt{L_3 \cdot C_1}$ In the region of Mode II	DC-DC boost Input: 20–45 V dc Output: 54 Vdc, 270 W F_{sw} : 95 kHz η : 97 % at 24 Vdc, full load	✓ Soft switching on both turn-on and turn-off ✓ Small extra inductors needed ✓ Less cost and volume from low- rating components ✓ High efficiency when low input voltage or light load ✓ One snubber components located on main power paths ✗ Two extra inductors used

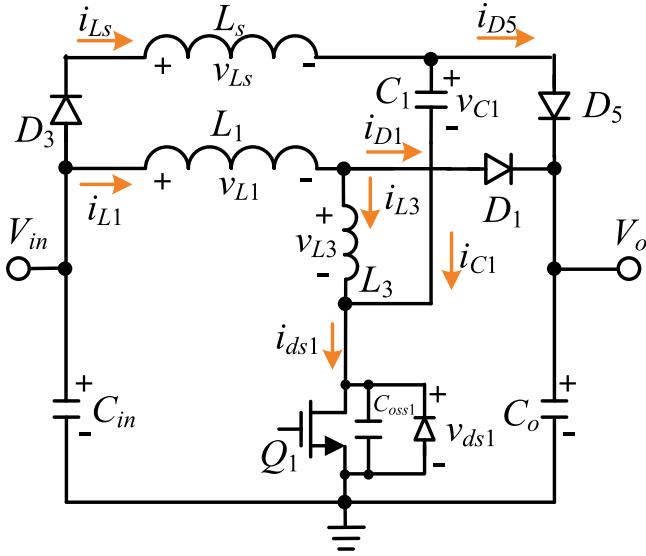


Fig. 2. The proposed non-dissipative snubber.

2) State 2 ($t_1 - t_2$): As displayed in Fig. 5(b), Q_1 is kept in the on-state. In this state, L_1 is still magnetized, causing i_{L1} to increase. Because L_1 is much higher than L_3 , L_3 can be ignored to obtain (24). Moreover, i_{D1} is 0. L_s and C_1 resonate, causing v_{C1} to increase. i_{Ls} and v_{C1} are expressed by (25) and (27), respectively. When i_{Ls} is zero, the converter transitions to State 3.

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_1) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_1) \approx i_{L1}(t_1) + \frac{V_{in}}{L_1} (t - t_1) \quad (24)$$

$$i_{C1}(t) = i_{Ls}(t) = i_{Ls}(t_1) \cdot \cos \omega_0 (t - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0} \right) \cdot \sin \omega_0 (t - t_1) \quad (25)$$

$$\begin{aligned} i_{C1}(t_2) &= i_{Ls}(t_2) \\ &= i_{Ls}(t_1) \cdot \cos \omega_0 (t_2 - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0} \right) \cdot \sin \omega_0 (t_2 - t_1) = 0 \end{aligned} \quad (26)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0 (t - t_1) \quad (27)$$

$$v_{C1}(t_2) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0 (t_2 - t_1) = 2 \cdot V_{in} - v_{C1}(t_0) = v_{C1}(\text{PPK-I}) \quad (28)$$

$$\Delta t_2 = (t_2 - t_1) = \frac{T_0}{2} - t_1 \quad (29)$$

3) State 3 ($t_2 - t_3$): As displayed in Fig. 5(c), Q_1 is kept in the on-state, causing i_{L1} to increase continuously. The currents in L_s , C_1 , and D_5 are zero. When Q_1 is turned off, the converter transitions to State 4.

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_2) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_2) = i_{L1}(t_2) + \frac{V_{in}}{L_1} (t - t_2) \quad (30)$$

$$i_{L1}(t_3) = i_{L3}(t) = i_{L1}(t_2) + \frac{V_{in}}{L_1} (t_3 - t_2) = i_{L1-pk} \quad (31)$$

$$v_{C1}(t) = v_{C1}(t_2) = v_{C1}(\text{PPK-I}) \quad (32)$$

$$\Delta t_3 = (t_3 - t_2) = T_{on} - t_2 \quad (33)$$

4) State 4 ($t_3 - t_4$): As displayed in Fig. 5(d), Q_1 is turned off. The freewheeling currents i_{L1} and i_{L3} discharge C_1 , causing D_5 to become forward-biased. As a result, v_{ds1} gradually increases, reducing the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . The turn-off switching loss is

Table 2

Definition list of symbols and variables.

Symbol and variable	Definition
V_{in}, V_o, P_o	Input voltage, output voltage, output power, respectively
I_{in}	Average of i_{in}
$I_{o, rated}$	Rated load current of boost stage
η_{bst}, η_{PSU}	Expected efficiency of boost stage and PSU, respectively
L_1	Main inductor of boost stage
L_s, L_3	Snubber inductors of boost stage
Q_1	Main switch
C_{in}, C_o	Input capacitor and output capacitor, respectively
D_1	Output rectifier diode
C_{oss1}	Output capacitance of Q_1
L_s, L_3	Resonance inductors of snubber
C_1	Resonance capacitor of snubber
D_3, D_5	Diode of snubber
f_{sw}, T_{sw}	Switching frequency and switching period, respectively
D	Duty cycle
T_{on}	Conduction time of Q_1
v_{L1}	Voltage of L_1
i_{L1}	Current of L_1 , equal to i_{D1}
Δi_{L1}	Ripple current of L_1
i_{L1-vfy}	Equal to $i_{L1-pk} - \Delta i_{L1}$
i_{Ls}, v_{Ls}	Current and voltage of L_s , respectively
i_{ds1}, v_{ds1}	Drain-source current and voltage of Q_1
i_{ds1-pk}, v_{ds1-pk}	Drain-source peak current and voltage of Q_1
i_{L3}, v_{L3}	Current and voltage of L_3 , respectively
i_{Co-rms}	Root mean square current of C_o
i_{C1}, v_{C1}	Current and voltage of C_1
$i_{L1-pk}, i_{Ls-pk}, i_{L3-pk}$	Peak currents of L_1, L_s , and L_3
$v_{C1}(\text{NPK-I}), v_{C1}(\text{PPK-I})$	Minimum and Maximum voltages of C_1 in Mode I
$v_{C1}(\text{NPK-II}), v_{C1}(\text{PPK-II})$	Minimum and Maximum voltages of C_1 in Mode II
$v_{C1}(\text{NPK-III}), v_{C1}(\text{PPK-III})$	Minimum and Maximum voltages of C_1 in Mode III
$v_{C1}(\text{NPK-IV}), v_{C1}(\text{PPK-IV})$	Minimum and Maximum voltages of C_1 in Mode IV
$i_{L1-rms}, i_{Ls-rms}, i_{L3-rms}$	RMS values of i_{L1}, i_{Ls} , and i_{L3} , respectively
$V_{F-D1}, V_{F-D3}, V_{F-D5}$	Forward voltages of D_1, D_3 , and D_5 , respectively
i_{D1}, i_{D3}, i_{D5}	Currents of D_1, D_3 , and D_5 , respectively
$i_{D1-avg}, i_{D3-avg}, i_{D5-avg}$	Average currents of D_1, D_3 , and D_5 , respectively
$i_{D1-pk}, i_{D3-pk}, i_{D5-pk}$	Peak currents of D_1, D_3 , and D_5 , respectively
i_{D1-rms}	RMS value of i_{D1}
Z_0	Impedance of resonant tank of L_s and C_1
Z_1	Impedance of resonant tank of L_1 and C_1
Z_2	Impedance of resonant tank of L_3 and C_1
ω_0	Angular frequency of resonant tank of L_s and C_1
ω_1	Angular frequency of resonant tank of L_1 and C_1
ω_2	Angular frequency of resonant tank of L_3 and C_1
T_0	Resonant period of L_s and C_1
T_1	Resonant period of L_1 and C_1
T_2	Resonant period of L_3 and C_1
f_0	Resonant frequency of L_s and C_1
$V_{in(\min)}$	Minima input voltage
D_{max}	Maxima duty cycle
i_{L1-avg}, i_{L1-rms}	Average and RMS value of i_{L1}
N_{L1}	Winding turns of L_1
A_{L1}	Inductance factor of L_1
A_{e-L1}	Effective magnetic cross-section of L_1
V_{e-L1}	Effective magnetic volume of L_1
B_s, B_{max}	Saturation and maximum magnetic flux density of L_1 , respectively
$\text{CoreLoss}_{(380\text{mT}, 100\text{ kHz})}$	Core loss of L_1 with unit volume
$v_{C1}(\text{NPK}), v_{C1}(\text{PPK})$	Negative peak and positive peak voltages of C_1 under all modes, respectively
Q_{g1}	Gate charge of Q_1
V_{DR1}	Maxima v_{gs1} of Q_1
V_{th1}	Threshold voltage of Q_1
$R_{g1(\text{int})}$	Internal gate resistance of Q_1
$R_{g1(\text{on-ext}), R_{g1(\text{off-ext})}$	External resistors for turn on and turn off of Q_1
$t_{Q1\text{-turn-on}}, t_{Q1\text{-turn-off}}$	Calculated rise and fall times of v_{ds1} under hard switching
t_{r-vds1}	Calculated rise time of v_{ds1} when Q_1 turned-off with proposed snubber
$t_{L3\text{-turn-on}}$	Calculated rise time of i_{L3} and i_{ds1} when Q_1 turned-on with proposed snubber
$R_{ds1(\text{on})}$	Drain-source on-resistance of Q_1
R_{L1}, R_{Ls}, R_{L3}	DC resistance of L_1, L_s , and L_3 , respectively
R_{C1}, R_{Co}	ESR resistance of C_1 and C_o , respectively

(continued on next page)

Table 2 (continued)

Symbol and variable	Definition
L_{C1}	ESL inductance of C_1
Q_{rr-D1}	Reverse recovery charge of D_1
$P_{L1-core}, P_{L1-copper}$	Core and copper losses of L_1 , respectively
P_{Ls}, P_{L3}	Losses of L_s and L_3 , respectively
$P_{Q1-sw}, P_{Q1-dr}, P_{Q1-con}$	Switching loss, driving loss, and conduction loss of Q_1 , respectively
P_{Q1-tot}	Total loss of Q_1
$P_{D1}, P_{D3}, P_{D5}, P_{Co}$	Losses of D_1, D_3, D_5, C_o , respectively
P_{tot}	Total loss of boost stage

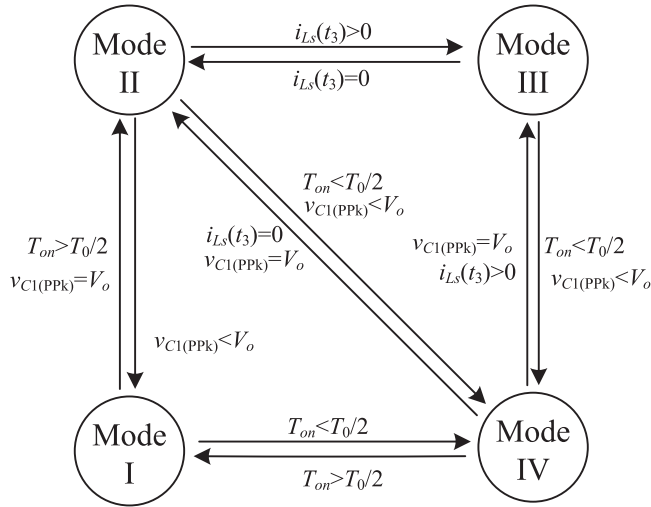


Fig. 3. State machine and operation modes of the proposed circuit.

reduced. The maximum currents of i_{D5} and i_{C1} are equal to i_{L1-pk} . As C_1 gradually discharges from $v_{C1}(PPK-I)$, v_{C1} can be expressed as (34). As L_1 is significantly higher than L_3 , v_{L1} is much higher than v_{L3} . As a result, v_{L3} is treated as zero. Then, $v_{L1}(t)$ can be simplified to (36). By assuming that L_1 can be considered as a current source for a short period, the rise time of v_{ds1} can be obtained from (39). When $v_{C1}(t_4)$ (equal to $v_{D1}(t_4)$) is zero, the converter transitions to State 5.

$$v_{C1}(t) = v_{C1}(t_3) + \left(\frac{-i_{L1}(t_3) \cdot (t - t_3)}{C_1} \right) = v_{C1}(PPK-I) - \frac{i_{L1-pk} \cdot (t - t_3)}{C_1} \quad (34)$$

$$v_{C1}(t_4) = V_o - \frac{i_{L1-pk}(t_4 - t_3)}{C_1} = 0 \quad (35)$$

$$v_{L1}(t) = V_o - V_{in} + v_{C1}(t) \quad (36)$$

$$\begin{aligned} i_{L1}(t) &= i_{C1}(t) = i_{L3}(t) = i_{D5}(t) \\ &= i_{L1}(t_3) \cdot \cos \omega_1(t - t_3) + \left(\frac{V_{in} + v_{C1}(t) - V_o}{Z_1} \right) \cdot \sin \omega_1(t - t_3) \end{aligned} \quad (37)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_3) \quad (38)$$

$$\Delta t_4 = (t_4 - t_3) = \frac{C_1 \cdot (v_{C1}(PPK-I) - 0)}{i_{L1-pk}} \quad (39)$$

5) State 5 (t_4 – t_5): As displayed in Fig. 5(e), Q_1 is kept in the off-state. Freewheeling current i_{L3} causes D_5 to be forward-biased and transfer its energy to C_1 . Since L_3 resonates with C_1 , i_{L3} and v_{C1} can be expressed as (41) and (44), respectively. When i_{L3} reduces to zero, v_{C1} reaches its negative peak value. The peak voltage of v_{ds1} is represented as (46). Simultaneously, the converter transitions to State 6.

$$i_{L1}(t) = i_{L1}(t_4) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_4) \quad (40)$$

$$\begin{aligned} i_{L3}(t) &= i_{D5}(t) = -i_{C1}(t) \\ &= i_{L1-pk} \cdot \cos \omega_2(t - t_4) + \frac{v_{C1}(t_4)}{Z_2} \cdot \sin \omega_2(t - t_4) \end{aligned} \quad (41)$$

$$i_{L3}(t_5) = i_{D5}(t_5) = -i_{C1}(t_5) = 0 \quad (42)$$

$$v_{C1}(t) = 0 + Z_2 \cdot i_{C1}(t_4) \cdot \sin \omega_2(t - t_4) \quad (43)$$

$$v_{C1}(t_5) = -\sqrt{\frac{(-i_{L1-pk})^2 \cdot L_3}{C_1}} = v_{C1(NPK-I)} \quad (44)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_4) \quad (45)$$

$$v_{ds1}(t_5) = V_o - v_{C1(NPK-I)} \quad (46)$$

$$\Delta t_5 = (t_5 - t_4) = \frac{T_2}{4} \quad (47)$$

6) State 6 (t_5 – $t_0 + T_{SW}$): As displayed in Fig. 5(f), Q_1 is kept in the off-state and D_1 is kept in the on-state. In this condition, there is no current flowing through any of the components of the snubber. Simultaneously, the energy stored in L_1 is transferred to the output. Hence, i_{L1} decreases gradually. When Q_1 is turned off, the converter returns to Stage 1.

$$i_{L1}(t) = i_{D1}(t) = i_{L1}(t_5) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_5) \quad (48)$$

$$i_{C1}(t) = 0 \quad (49)$$

$$v_{C1}(t) = v_{C1}(t_5) \quad (50)$$

$$v_{ds1}(t) = V_o \quad (51)$$

$$\Delta t_6 = (t_0 + T_{SW} - t_5) = T_{SW} - t_5 \quad (52)$$

2.2. Operation mode II

The simulation waveforms of Mode II are displayed in Fig. 6. First, there are some initial conditions, as displayed in (12)–(15).

1) State 1 (t_0 – t_1): As displayed in Fig. 7(a), Q_1 is turned on. The initial value of i_{L1} is expressed by (10). In this state, i_{L3} increases from zero but is still less than i_{L1} , indicating that there is a partial current flowing through D_1 (termed i_{D1}), as expressed in (53). L_1 is demagnetized to decrease i_{L1} . V_{in} charges the resonant tank of L_s and C_1 . i_{Ls} and v_{C1} can be expressed as (54) and (57), respectively. Simultaneously, i_{L3} and i_{ds1} start to increase from zero, as displayed in (55) and (58), to decrease the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . As a result, the turn-on switching loss is reduced. The rise time of i_{ds1} is calculated as shown in (60). However, i_{L3} is still less than i_{L1} . This means there is a partial current flowing through D_1 (termed i_{D1}), which is expressed in (59). When i_{D1} reduces to zero, the converter enters State 2.

$$i_{L1}(t) = i_{L1}(t_0) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_0) \quad (53)$$

$$i_{Ls}(t) = i_{C1}(t) = i_{Ls}(t_0) + \left(\frac{V_{in} - v_{C1}(t_0)}{Z_0} \right) \cdot \sin \omega_0(t - t_0) \quad (54)$$

$$i_{L3}(t) = i_{L3}(t_0) + \left(\frac{V_o - 0}{L_3} \right) \cdot (t - t_0) = \left(\frac{V_o - 0}{L_3} \right) \cdot (t - t_0) \quad (55)$$

$$i_{L3}(t_1) = i_{L1}(t_1) \quad (56)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_0)) \cdot \cos \omega_0(t - t_0) \quad (57)$$

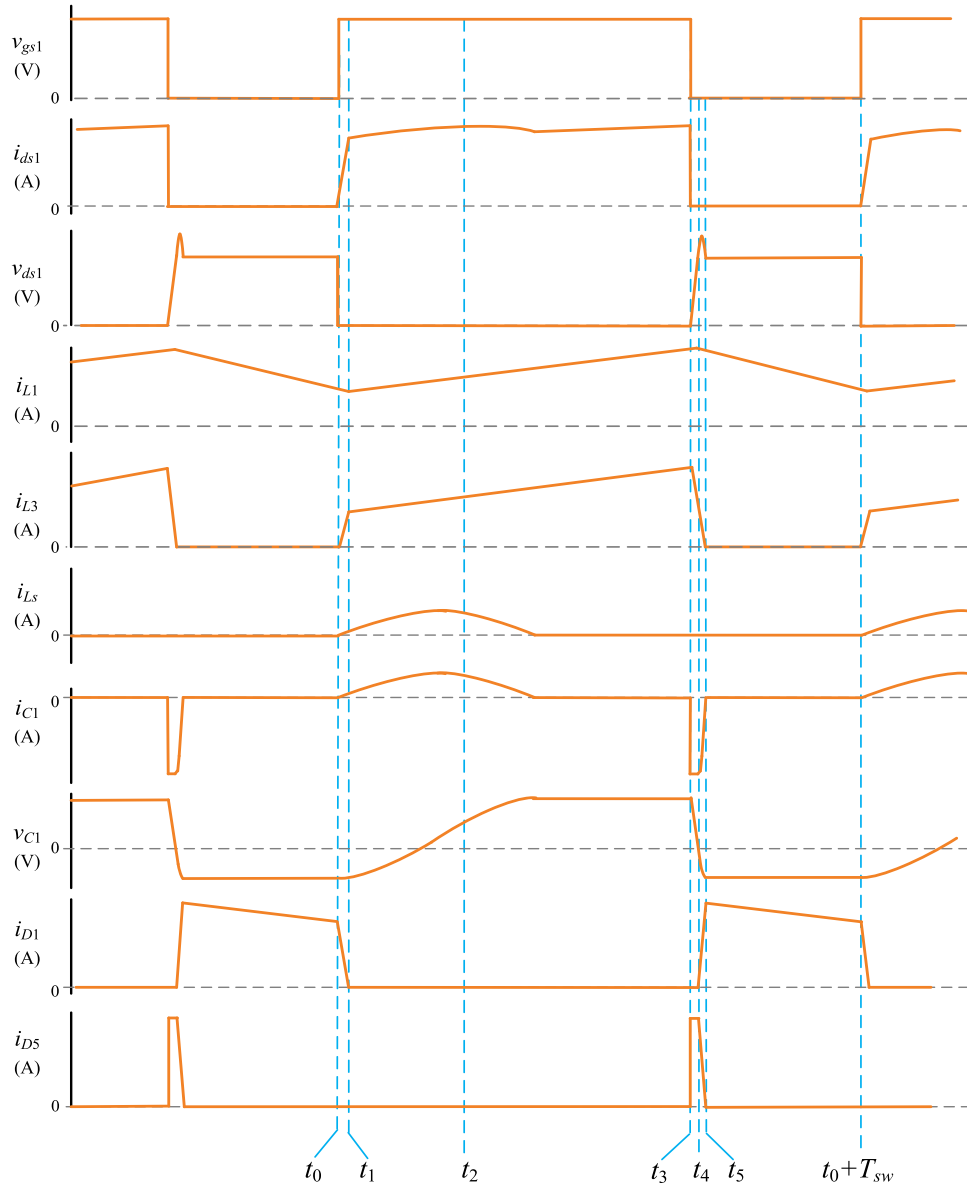


Fig. 4. Simulation waveforms under Mode I.

$$i_{ds1}(t) = i_{Ls}(t) + i_{L3}(t) \quad (58)$$

$$i_{D1}(t) = i_{L1}(t) - i_{L3}(t) \quad (59)$$

$$\Delta t_1 = (t_1 - t_0) = \frac{L_3 \cdot i_{L1}(t_0)}{V_o} \quad (60)$$

2) State 2 (t_1 – t_2): As displayed in Fig. 7(b), Q_1 is kept in the on-state. In this state, L_1 remains magnetized, causing i_{L1} to increase. Because L_1 is significantly higher than L_3 , L_3 can be ignored to rewrite (61) as (62). Further, i_{D1} is zero, L_s and C_1 resonate to make v_{C1} increase, and i_{Ls} and v_{C1} are expressed by (63) and (64), respectively. When $v_{C1}(t_2)$ is charged to V_o , the converter transitions to State 3.

$$i_{L1}(t) = i_{L1}(t_1) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_1) \quad (61)$$

$$i_{L1}(t) = i_{L1}(t_1) + \frac{V_{in}}{L_1} (t - t_1) \quad (62)$$

$$i_{C1}(t) = i_{Ls}(t) = i_{Ls}(t_1) \cdot \cos \omega_0 (t - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0} \right) \cdot \sin \omega_0 (t - t_1) \quad (63)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0 (t - t_1) \quad (64)$$

$$v_{C1}(t_2) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0 (t_2 - t_1) = V_o = v_{C1}(\text{PPK-II}) \quad (65)$$

$$\Delta t_2 = (t_2 - t_1) = \frac{\arccos \left(\frac{V_{in} - V_o}{V_{in} - v_{C1}(t_1)} \right)}{\omega_0} \quad (66)$$

3) State 3 (t_2 – t_3): As displayed in Fig. 7(c), Q_1 is kept in the on-state and L_1 remains magnetized to cause i_L to increase. Because $v_{C1}(t_2)$ is charged to V_o , the freewheeling current i_{Ls} causes D_5 to be forward-biased. i_{Ls} is expressed as (68). When $i_{Ls}(t_3)$ is demagnetized to zero, the converter transitions to State 4.

$$i_{L1}(t) = i_{L1}(t_2) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_2) = i_{L1}(t_2) + \frac{V_{in}}{L_1} (t - t_2) \quad (67)$$

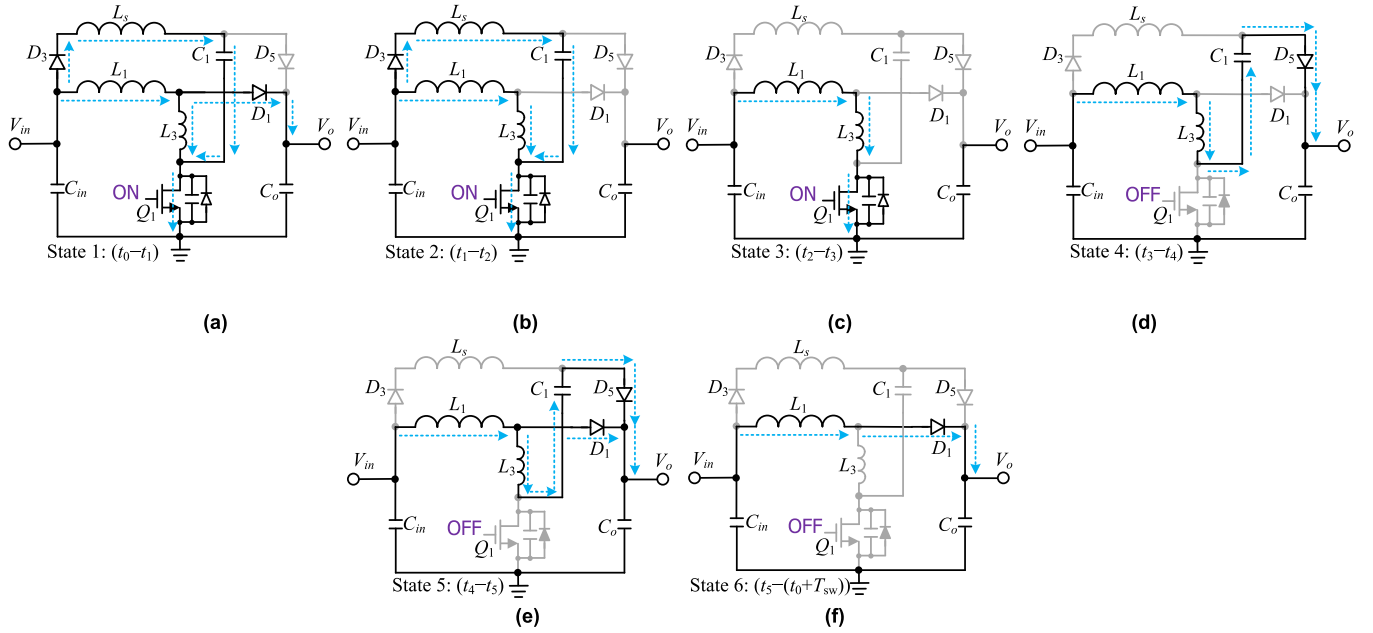


Fig. 5. Operation states and their equivalent models of Mode I: (a) State 1, (b) State 2, (c) State 3, (d) State 4, (e) State 5, and (f) State 6.

$$i_{Ls}(t) = i_{Ls}(t_2) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t - t_2) \quad (68)$$

$$i_{Ls}(t_3) = i_{Ls}(t_2) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t_3 - t_2) = 0 \quad (69)$$

$$v_{C2}(t_3) = V_o = v_{C1(PPK-II)} \quad (70)$$

$$i_{ds1}(t) = i_{L1}(t - t_2) \quad (71)$$

$$\Delta t_3 = (t_3 - t_2) = \frac{L_s \cdot (0 - i_{Ls}(t_2))}{V_{in} - V_o} = \frac{L_s \cdot i_{Ls}(t_2)}{V_o - V_{in}} \quad (72)$$

4) State 4 (t_3-t_4): As displayed in Fig. 7(d), Q_1 is kept in the on-state, causing i_{L1} to increase continuously. The currents in L_s , C_1 , and D_1 are 0. When Q_1 is turned off, the converter transitions to State 5.

$$i_{L1}(t) = i_{L1}(t_3) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_3) = i_{L1}(t_3) + \frac{V_{in}}{L_1} (t - t_3) \quad (73)$$

$$i_{L1}(t_4) = i_{L1}(t_3) + \frac{V_{in}}{L_1} (t_4 - t_3) = i_{L1-pk} \quad (74)$$

$$v_{C1}(t) = V_o \quad (75)$$

$$\Delta t_4 = (t_4 - t_3) = T_{on} - t_3 \quad (76)$$

5) State 5 (t_4-t_5): As displayed in Fig. 7(e), Q_1 is turned off. The freewheeling currents i_{L1} and i_{L3} discharge C_1 , causing D_5 to become forward-biased. As a result, v_{ds1} gradually increases, reducing the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . The turn-off switching loss is reduced. The maximum currents of i_{D5} and i_{C1} are equal to i_{L1-pk} . As C_1 gradually discharges from V_o , v_{C1} can be expressed as (77). L_1 is significantly higher than L_3 , resulting in v_{L1} being much higher than v_{L3} . Therefore, v_{L3} is treated as 0. By assuming that L_1 can be considered a current source for a short period, the rise time of v_{ds1} and the time required for State 5 can be obtained from (81). When both $v_{D1}(t)$ and $v_{C1}(t)$ are 0, it is ensured that $i_{D1}(t)$ is greater than 0, and the converter transitions to State 6.

$$v_{C1}(t) = v_{C1}(t_4) + \left(\frac{-i_{L1}(t_4) \cdot (t - t_4)}{C_1} \right) = V_o - \frac{i_{L1-pk} \cdot (t - t_4)}{C_1} \quad (77)$$

$$v_{C1}(t_5) = V_o - \frac{i_{L1-pk}(t_5 - t_4)}{C_1} = 0 \quad (78)$$

$$\begin{aligned} i_{L1}(t) &= i_{C1}(t) = i_{D5}(t) \\ &= i_{L1}(t_4) \cdot \cos \omega_1(t - t_4) + \left(\frac{V_{in} + v_{C1}(t_4)}{Z_1} - V_o \right) \cdot \sin \omega_1(t - t_4) \end{aligned} \quad (79)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_4) \quad (80)$$

$$\Delta t_5 = (t_5 - t_4) = \frac{C_1 \cdot (V_o - 0)}{i_{L1-pk}} \quad (81)$$

6) State 6 (t_5-t_6): As displayed in Fig. 7(f), Q_1 is kept in the off-state. Freewheeling current i_{L3} makes D_5 forward-biased and transfers the energy to C_1 . Due to the resonance between L_3 and C_1 , the expressions for i_{L3} , and v_{C1} can be described by (83) and (85), respectively. When i_{L3} reduces to zero, v_{C1} reaches its negative peak value. The peak voltage of v_{ds1} is represented as (88). Simultaneously, the converter transitions to State 7.

$$i_{L1}(t) = i_{L1}(t_5) + \left(\frac{V_{in} - V_o}{L_1} \right) \cdot (t - t_5) \quad (82)$$

$$i_{L3}(t) = i_{D5}(t) = -i_{C1}(t) = i_{L1-pk} \cdot \cos \omega_2(t - t_5) + \frac{v_{C1}(t_5)}{Z_2} \cdot \sin \omega_2(t - t_5) \quad (83)$$

$$i_{L3}(t_6) = i_{D5}(t_6) = -i_{C1}(t_6) = 0 \quad (84)$$

$$v_{C1}(t) = 0 - (0 - v_{C1}(t_5)) \cdot \cos \omega_2(t - t_5) + Z_2 \cdot i_{C1}(t_5) \cdot \sin \omega_2(t - t_5) \quad (85)$$

$$v_{C1}(t_6) = -\sqrt{\frac{(-i_{L1-pk})^2 \cdot L_3}{C_1}} = v_{C1(NPK-II)} \quad (86)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_5) \quad (87)$$

$$v_{ds1}(t_6) = V_o - v_{C1}(t_6) = V_o - v_{C1(NPK-II)} \quad (88)$$

$$i_{D1}(t) = i_{L1}(t) - i_{L3}(t) \quad (89)$$

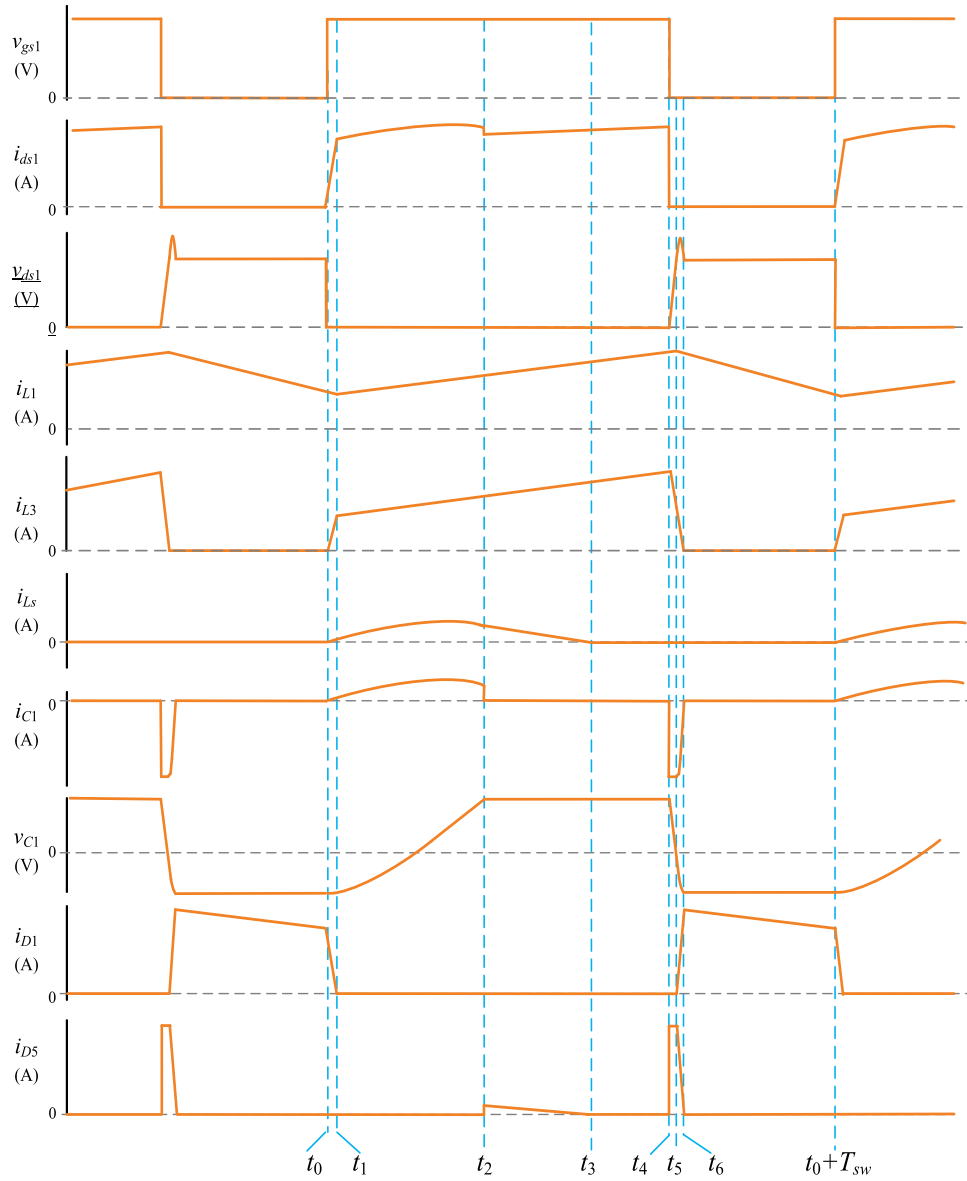


Fig. 6. Simulations waveforms under Mode II.

$$\Delta t_6 = (t_6 - t_5) = \frac{T_2}{4} \quad (90)$$

7) State 7 ($t_6 - t_0 + T_{SW}$): As displayed in Fig. 7(g), Q_1 is kept in the off-state and D_1 is kept in the on-state. In this condition, there is no current flowing through any of the components of the snubber. Simultaneously, the energy stored in L_1 will be transferred to the output. Hence, i_{L1} gradually decreases. When Q_1 is turned on, the converter returns to State 1.

$$i_{L1}(t) = i_{D1}(t) = i_{L1}(t_6) + \frac{(V_{in} - V_o)}{L_1}(t - t_6) \quad (91)$$

$$v_{C1}(t_0 + T_{SW}) = v_{C1}(t_6) = v_{C1(NPk-II)} \quad (92)$$

$$v_{ds1}(t) = V_o \quad (93)$$

$$\Delta t_6 = (t_0 + T_{SW} - t_6) = T_{SW} - t_6 \quad (94)$$

2.3. Operation mode III

The simulation waveforms of Mode III are displayed in Fig. 8. First, there are some initial conditions, as displayed in (12)–(14) and (95).

$$v_{C1}(t_0) = \left(V_o - V_{in} \right) - \sqrt{\frac{(i_{L1-pk})^2 \cdot L_3}{C_1}} \quad (95)$$

1) State 1 ($t_0 - t_1$): As displayed in Fig. 9(a), Q_1 is turned on. The initial value of i_{L1} is described in (10). In this state, i_{L3} increases from zero but is still less than i_{L1} , indicating that there is a partial current flowing through D_1 (termed i_{D1}), as expressed in (96). L_1 is demagnetized to decrease i_{L1} . V_{in} charges the resonant tank of L_s and C_1 . i_{Ls} can be expressed as (97). Simultaneously, i_{L3} and i_{ds1} start to increase from zero, as displayed in (99) and (102), to decrease the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . As a result, the turn-on switching loss is reduced. The rise time of i_{ds1} is calculated as shown in (104). However, i_{L3} is still less than i_{L1} , meaning there is a partial current flowing through D_1 (termed i_{D1}), which is expressed as (103). v_{C1} can be expressed as (101). When i_{D1} reduces to zero, the converter transitions to State 2.

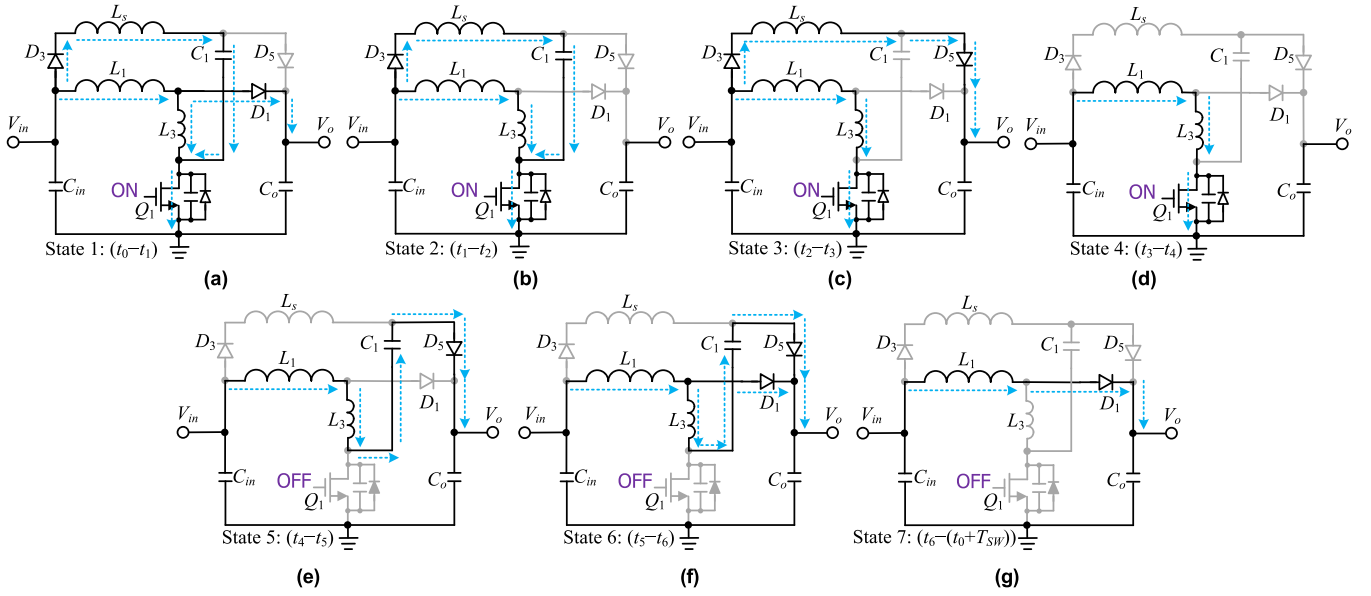


Fig. 7. Operation states and their equivalent models of Mode II: (a) State 1, (b) State 2, (c) State 3, (d) State 4, (e) State 5, (f) State 6, and (g) State 7.

$$i_{L1}(t) = i_{L1}(t_0) + \left(\frac{V_{in} - V_o}{L_1 + L_3} \right) \cdot (t - t_0) = i_{L1}(t_0) + \left(\frac{V_{in} - V_o}{L_1} \right) \cdot (t - t_0) \quad (96)$$

$$i_{Ls}(t) = i_{C1}(t) = i_{Ls}(t_0) \cdot \cos \omega_0(t - t_0) + \left(\frac{V_{in} - v_{C1}(t_0)}{Z_0} \right) \cdot \sin \omega_0(t - t_0) \quad (97)$$

$$i_{Ls}(t_1) = i_{C1}(t_1) = \left(\frac{V_{in} - v_{C1}(t_0)}{Z_0} \right) \cdot \sin \omega_0(t_1 - t_0) \quad (98)$$

$$i_{L3}(t) = i_{L3}(t_0) + \left(\frac{V_o - 0}{L_3} \right) \cdot (t - t_0) = \left(\frac{V_o - 0}{L_3} \right) \cdot (t - t_0) \quad (99)$$

$$i_{L3}(t_1) = i_{L1}(t_1) \quad (100)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_0)) \cdot \cos \omega_0(t - t_0) \quad (101)$$

$$i_{ds1}(t) = i_{Ls}(t) + i_{L3}(t) \quad (102)$$

$$i_{D1}(t) = i_{L1}(t) - i_{L3}(t) \quad (103)$$

$$\Delta t_1 = (t_1 - t_0) = \frac{L_3 \cdot i_{L1}(t_0)}{V_o} \quad (104)$$

2) State 2 (t_1-t_2): As displayed in Fig. 9(b), Q_1 is kept in the on-state. In this state, L_1 is still magnetized, causing i_{L1} to increase, and i_{D1} is zero. Since L_1 is significantly higher than L_3 , it is advisable to ignore v_{L3} when L_1 and L_3 are cascaded. $i_{L1}(t)$ is written as (105). L_s and C_1 resonate to make v_{C1} increase, and i_{Ls} and v_{C1} are expressed by (106) and (107), respectively. When $v_{C1}(t_2)$ is charged to V_o , the converter transitions to State 3.

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_1) + \left(\frac{V_{in}}{L_1 + L_3} \right) (t - t_1) \approx i_{L1}(t_1) + \frac{V_{in}}{L_1} (t - t_1) \quad (105)$$

$$i_{C1}(t) = i_{Ls}(t) = i_{Ls}(t_1) \cdot \cos \omega_0(t - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0} \right) \cdot \sin \omega_0(t - t_1) \quad (106)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0(t - t_1) + Z_0 \cdot i_{C1}(t_1) \cdot \sin \omega_0(t - t_1) \quad (107)$$

$$v_{C1}(t_2) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0(t_2 - t_1) + Z_0 \cdot i_{C1}(t_1) \cdot \sin \omega_0(t_2 - t_1) = V_o = v_{C1}(\text{PPK-III}) \quad (108)$$

$$i_{ds1}(t) = i_{L1}(t) + i_{L2}(t) \quad (109)$$

$$\Delta t_2 = \left(t_2 - t_1 \right) = \frac{\arccos \left(\frac{V_{in} - V_o}{V_{in} - v_{C1}(t_1)} \right)}{\omega_0} \quad (110)$$

3) State 3 (t_2-t_3): As displayed in Fig. 9(c), Q_1 is kept in the on-state. L_1 is still magnetized to cause i_{L1} to increase. Because $v_{C1}(t_1)$ is charged to V_o , the freewheeling current i_{Ls} causes D_5 to become forward-biased. i_{Ls} is expressed by (112). When Q_1 is turned off, the converter transitions to State 4.

$$i_{L1}(t_3) = i_{L3}(t_3) = i_{L1}(t_2) + \frac{V_{in}}{L_1} (t_3 - t_2) = i_{L1-\text{pk}} \quad (111)$$

$$i_{Ls}(t) = i_{D5}(t) = i_{Ls}(t_2) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t - t_2) \quad (112)$$

$$i_{ds1}(t) = i_{L1}(t - t_2) \quad (113)$$

$$\Delta t_3 = (t_3 - t_2) = \frac{L_s \cdot (0 - i_{Ls}(t_2))}{V_{in} - V_o} = \frac{L_s \cdot i_{Ls}(t_2)}{V_o - V_{in}} \quad (114)$$

4) State 4 (t_3-t_4): As displayed in Fig. 9(d), Q_1 is turned off and the freewheeling currents i_{L1} and i_{L3} cause v_{C1} to increase. The freewheeling current of i_{Ls} is also demagnetized. Terms i_{L1} , i_{C1} , i_{Ls} , and v_{C1} can be expressed in (115), (116), and (117). During State 4, v_{C1} decreases from V_o to 0. As a result, v_{ds1} gradually increases, reducing the cross-sectional area of i_{ds1} and v_{ds1} . The turn-off switching loss is reduced. The rise time of v_{ds1} is calculated as shown in (120). When v_{C1} is equal to zero, the converter transitions to State 5.

$$i_{L1}(t) = i_{C1}(t) = i_{L3}(t) = i_{L1}(t_3) \cdot \cos \omega_1(t - t_3) + \left(\frac{V_{in} + v_{C1}(t_3) - V_o}{Z_1} \right) \cdot \sin \omega_1(t - t_3) \quad (115)$$

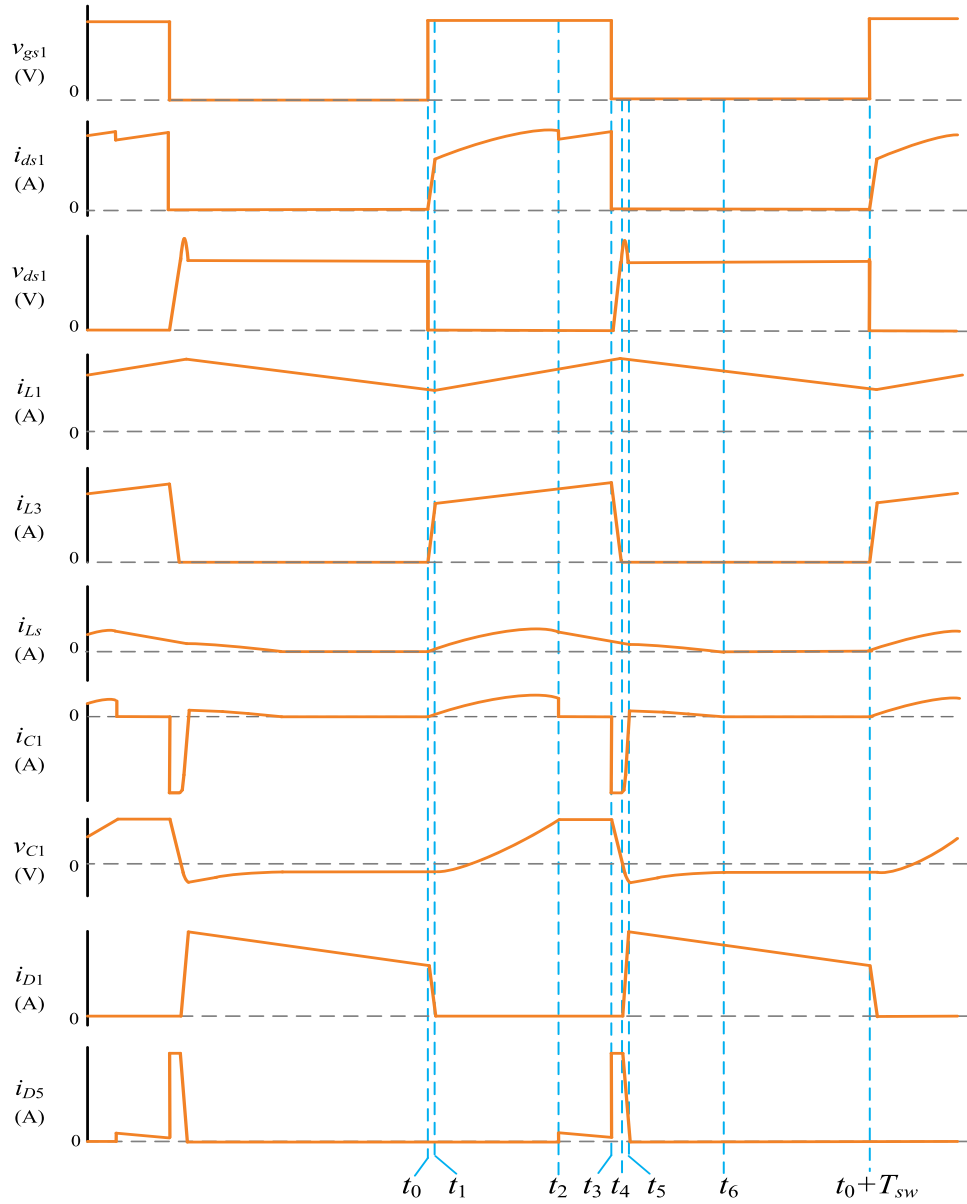


Fig. 8. Simulation waveforms under Mode III.

$$i_{Ls}(t) = i_{Ls}(t_3) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t - t_3) \quad (116)$$

$$v_{C1}(t) = v_{C1}(t_3) + \left(\frac{-i_{L1}(t_3)(t - t_3)}{C_1} \right) = V_o - \frac{i_{L1}(t_3)(t - t_3)}{C_1} \quad (117)$$

$$v_{C1}(t_4) = v_{C1}(t_3) + \left(\frac{-i_{L1}(t_3)(t_4 - t_3)}{C_1} \right) = V_o - \frac{i_{L1}(t_3)(t_4 - t_3)}{C_1} = 0 \quad (118)$$

$$\begin{aligned} i_{C1}(t_4) &= i_{L1}(t_4) \\ &= i_{L1}(t_3) \cdot \cos \omega_1(t_4 - t_3) + \left(\frac{V_{in} + v_{C1}(t_3) - V_o}{Z_1} \right) \cdot \sin \omega_1(t_4 - t_3) \end{aligned} \quad (119)$$

$$v_{ds1}(t) = V_o - v_{C1}(t) \quad (120)$$

$$i_{D5}(t) = i_{Ls}(t) + i_{L1}(t) \quad (121)$$

$$\Delta t_4 = (t_4 - t_3) = \frac{C_1 \cdot (V_o - 0)}{i_{L1-pk}} \quad (122)$$

5) State 5 (t_4 – t_5): As displayed in Fig. 9(e), Q_1 is kept in the off-state. Currents i_{L1} and i_{Ls} are continuously demagnetized. Because v_{C1} is zero, D_1 is forward-biased, causing i_{D1} to increase from 0. Terms i_{L1} , i_{Ls} , i_{C1} , and v_{C1} can be expressed in (123), (124), (125), and (128), respectively. During State 5, i_{L3} and i_{C1} decrease. When i_{L3} and i_{C1} reach zero, the converter transitions to State 6.

$$i_{L1}(t) = i_{L1}(t_4) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_4) \quad (123)$$

$$i_{Ls}(t) = i_{Ls}(t_4) + \left(\frac{V_{in} - V_o}{L_s} \right) (t - t_4) \quad (124)$$

$$i_{L3}(t) = i_{C1}(t) = i_{L1-pk} \cdot \cos \omega_2(t - t_4) + \frac{v_{C1}(t_4)}{Z_2} \cdot \sin \omega_2(t - t_4) \quad (125)$$

$$i_{L3}(t_5) = i_{C1}(t_5) = 0 \quad (126)$$

$$v_{C1}(t) = 0 - (0 - v_{C1}(t_4)) \cdot \cos \omega_1(t - t_4) + Z_2 \cdot i_{C1}(t_4) \cdot \sin \omega_2(t - t_4) \quad (127)$$

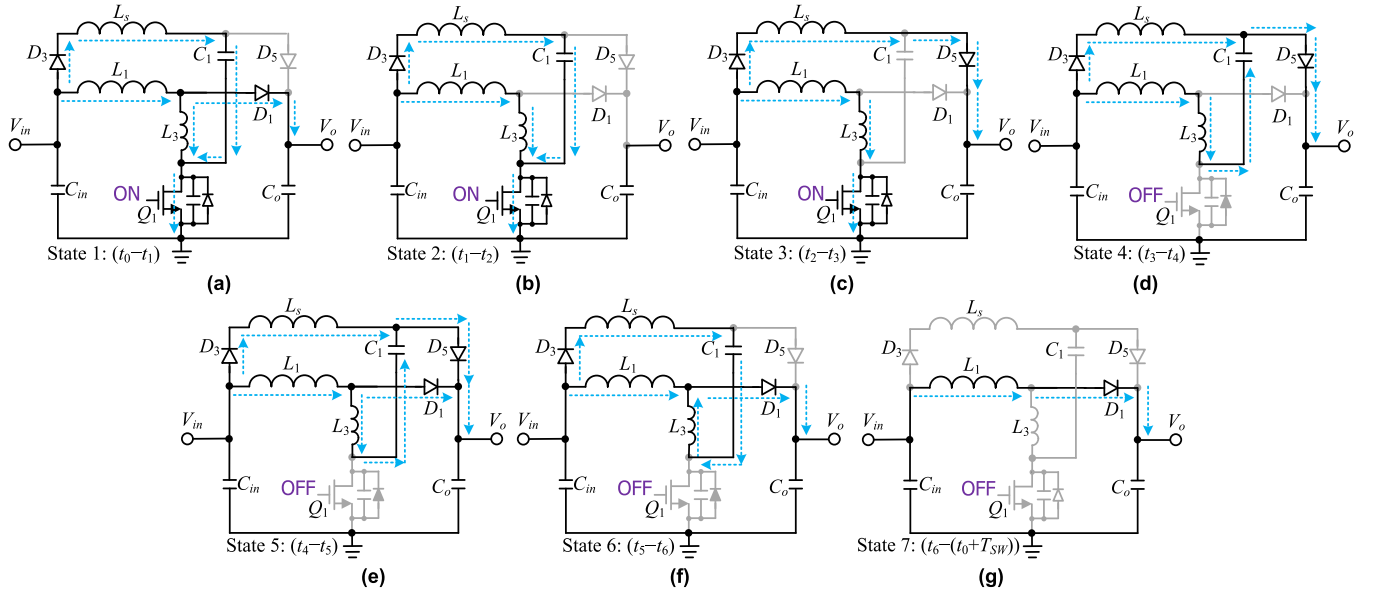


Fig. 9. Operation states and their equivalent models of Mode III: (a) State 1, (b) State 2, (c) State 3, (d) State 4, (e) State 5, (f) State 6, and (g) State 7.

$$v_{C1}(t_5) = -\sqrt{\frac{(-i_{L1-pk})^2 \cdot L_3}{C_1}} = v_{C1(NPk-III)} \quad (128)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_4) \quad (129)$$

$$v_{ds1}(t_5) = V_o - v_{C1}(t_5) = v_{ds1-pk} = V_o - v_{C1(NPk-III)} \quad (130)$$

$$i_{D5}(t_5) = i_{Ls}(t_5) + i_{C1}(t_5) \quad (131)$$

$$\Delta t_5 = (t_5 - t_4) = \frac{T_2}{4} \quad (132)$$

6) State 6 (t_5-t_6): As displayed in Fig. 9(f), Q_1 is kept in the off-state, while i_{L1} and i_{Ls} are continuously demagnetized. In this state, L_3 , L_s , and C_1 resonate. L_3 is magnetized reversely by V_{in} and v_{C2} . Since L_3 is significantly smaller than L_s , the inductance of L_3 and its voltage v_{L3} can be ignored when the cascaded L_s and L_3 resonate with C_1 . Therefore, the resonant tank can be described simply as C_1 resonating with L_s , i_{C1} is expressed in (134) and (135). v_{C1} is expressed in (137) and (138). When i_{C1} is discharged to zero, D_3 becomes reverse-biased and the converter transitions to State 7.

$$i_{L1}(t) = i_{L1}(t_5) + \left(\frac{V_{in} - V_o}{L_1}\right)(t - t_5) \quad (133)$$

$$\begin{aligned} i_{Ls}(t) &= -i_{L3}(t) = i_{C1}(t) \\ &= i_{Ls}(t_5) \cdot \cos \omega_0(t - t_5) + \left(\frac{V_{in} - V_o - v_{C1}(t)}{Z_0}\right) \cdot \sin \omega_0(t - t_5) \end{aligned} \quad (134)$$

$$i_{Ls}(t_6) = i_{C1}(t_6) = -i_{L3}(t_6) = 0 \quad (135)$$

$$i_{D1}(t) = i_{L1}(t) + i_{C1}(t) \quad (136)$$

$$\begin{aligned} v_{C1}(t) &= Z_0 \cdot i_{C1}(t_5) \cdot \sin \omega_0(t - t_5) + (V_{in} - V_o) - ((V_{in} - V_o) \\ &\quad - v_{C1}(t_5)) \cos \omega_0(t - t_5) \end{aligned} \quad (137)$$

$$v_{C1}(t_6) = v_{C1}(t_5) - (V_{in} - V_o) \quad (138)$$

$$v_{ds1}(t) = V_o - v_{C1}(t - t_5) \quad (139)$$

$$\Delta t_6 = (t_6 - t_5) = \frac{T_0}{2} \quad (140)$$

7) State 7 ($t_6-t_0+T_{SW}$): As displayed in Fig. 9(g), Q_1 is kept in the off-state, while D_1 is kept in the on-state. In this condition, there is no current flowing through any of the components of the snubber. Simultaneously, the energy stored in L_1 is transferred to the output. Hence, i_{L1} gradually decreases. The converter returns to State 1 when Q_1 is turned on.

$$i_{L1}(t) = i_{D1}(t) = i_{L1}(t_6) + \left(\frac{V_{in} - V_o}{L_1}\right)(t - t_6) \quad (141)$$

$$v_{C1}(t) = v_{C1}(t_6) \quad (142)$$

$$v_{ds1}(t) = V_o \quad (143)$$

$$\Delta t_7 = (t_0 + T_{SW} - t_6) = T_{SW} - t_6 \quad (144)$$

2.4. Operation mode IV

Fig. 10 displays the simulation waveforms under Mode IV when T_{on} is less than half a resonance period and $v_{C2} < V_o$. First, there are some initial conditions, as expressed in (12)–(14) and (145).

$$v_{C1}(t_0) = 0 \quad (145)$$

1) State 1 (t_0-t_1): As displayed in Fig. 11(a), Q_1 is turned on. The initial value of i_{L1} is described in (10). In this state, i_{L3} increases from zero but is still less than i_{L1} , indicating that there is a partial current flowing through D_1 (termed i_{D1}), as expressed in (146). L_1 is demagnetized to decrease i_{L1} . V_{in} charges the resonant tank of L_s and C_1 . i_{Ls} and v_{C1} can be expressed as (147) and (150), respectively. Simultaneously, i_{L3} and i_{ds1} start to increase from zero, as displayed in (148) and (152), to decrease the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . The turn-on switching loss is reduced. The rise time of i_{ds1} is calculated as shown in (154). However, i_{L3} is still less than i_{L1} , meaning there is a partial current flowing through D_1 (termed i_{D1}), which is expressed as (153). When i_{D1} reduces to zero, the converter transitions to State 2.

$$i_{L1}(t) = i_{L1}(t_0) + \left(\frac{V_{in} - V_o}{L_1}\right)(t - t_0) \quad (146)$$

$$i_{Ls}(t) = i_{C1}(t) = \left(\frac{V_{in} - v_{C1}(t_0)}{Z_0}\right) \cdot \sin \omega_0(t - t_0) \quad (147)$$

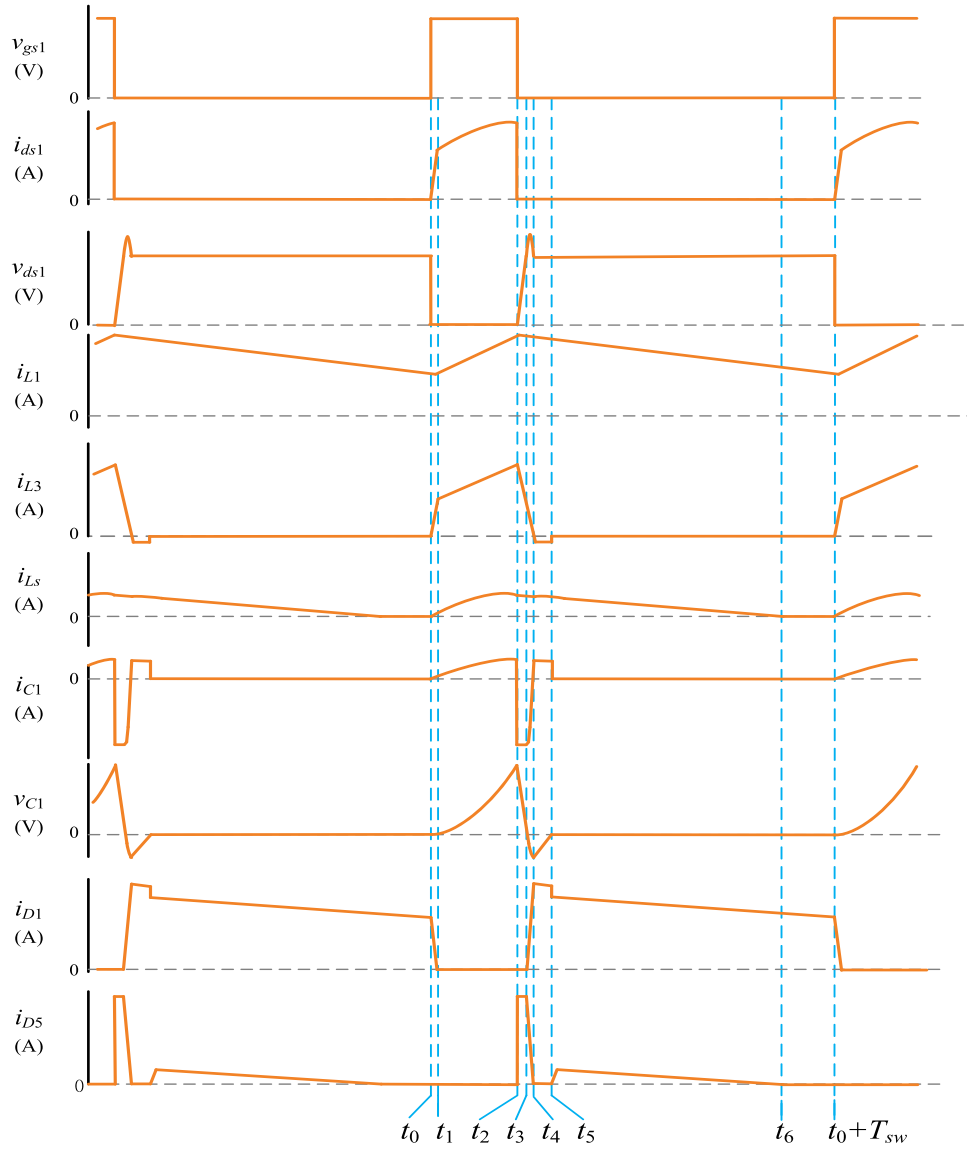


Fig. 10. Simulation waveforms under Mode IV.

$$i_{L3}(t) = i_{L3}(t_0) + \left(\frac{V_o - 0}{L_3}\right) \cdot (t - t_0) = \frac{V_o}{L_3} \cdot (t - t_0) \quad (148)$$

$$i_{L3}(t_1) = i_{L1}(t_1) \quad (149)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_0)) \cdot \cos \omega_0(t - t_0) + Z_0 \cdot i_{C1}(t_0) \cdot \sin \omega_0(t - t_0) \quad (150)$$

$$v_{C1}(t_1) = V_{in} - (V_{in} - 0) \cdot \cos \omega_0(t_1 - t_0) \quad (151)$$

$$i_{ds1}(t) = i_{Ls}(t) + i_{L3}(t) \quad (152)$$

$$i_{D1}(t) = i_{L1}(t) - i_{L3}(t) \quad (153)$$

$$\Delta t_1 = (t_1 - t_0) = \frac{L_3 \cdot i_{L1}(t_0)}{V_o} \quad (154)$$

2) State 2 (t_1 – t_2): As displayed in Fig. 11(b), Q_1 is kept in the on-state. In this state, L_1 is still magnetized, causing i_{L1} to increase. Since L_1 is significantly higher than L_3 , v_{L3} can be ignored when L_1 and L_3 are cascaded. Eq. (155) is rewritten as (156), and i_{D1} is 0. L_s and C_1 resonate to increase v_{C1} . i_{Ls} and v_{C1} are expressed as (158) and (160), respectively. When $v_{C1}(t_2)$ is charged to V_o , the converter transitions to State 3.

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_1) + \left(\frac{V_{in}}{L_1 + L_3}\right) (t - t_1) \quad (155)$$

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_1) + \frac{V_{in}}{L_1} (t - t_1) \quad (156)$$

$$i_{L1}(t_1) = i_{L3}(t_1) = i_{L1}(t_1) + \frac{V_{in}}{L_1} (t_2 - t_1) = i_{L1-pk} \quad (157)$$

$$i_{C1}(t) = i_{Ls}(t) = i_{Ls}(t) \cdot \cos \omega_0(t - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0}\right) \cdot \sin \omega_0(t - t_1) \quad (158)$$

$$\begin{aligned} i_{C1}(t_2) &= i_{Ls}(t_2) \\ &= i_{Ls}(t_1) \cdot \cos \omega_0(t_2 - t_1) + \left(\frac{V_{in} - v_{C1}(t_1)}{Z_0}\right) \cdot \sin \omega_0(t_2 - t_1) = 0 \end{aligned} \quad (159)$$

$$v_{C1}(t) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0(t - t_1) + Z_0 \cdot i_{Ls}(t) \cdot \sin \omega_0(t - t_1) \quad (160)$$

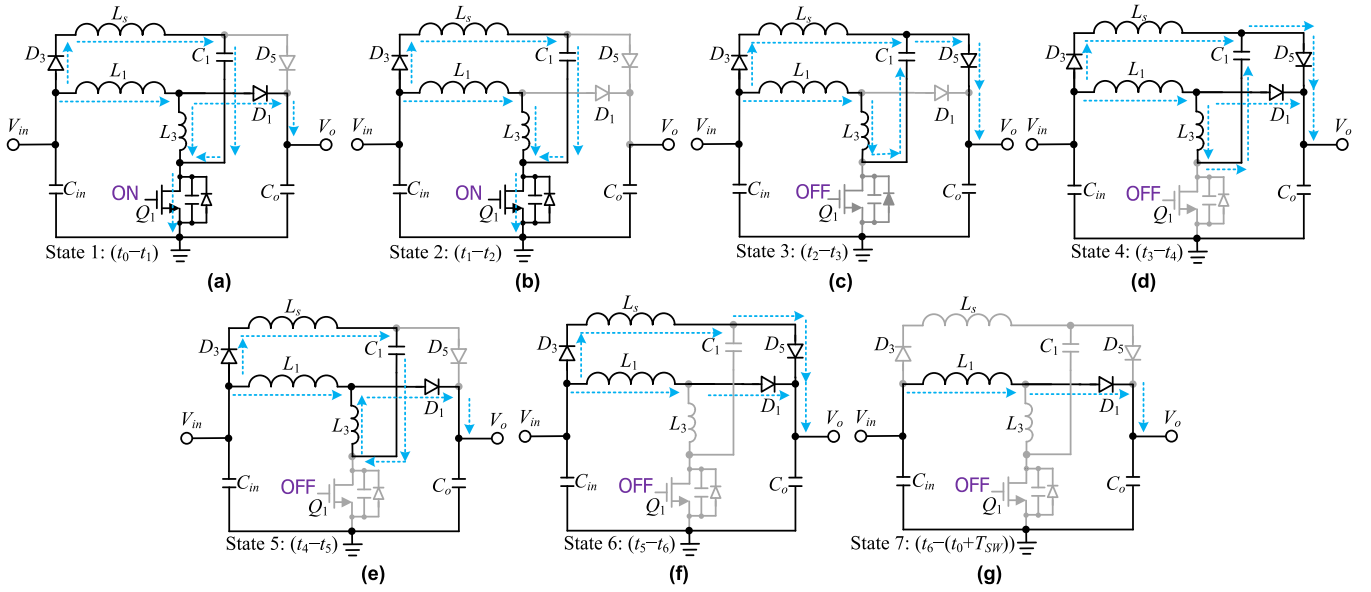


Fig. 11. Operation states and their equivalent models of Mode IV: (a) State 1, (b) State 2, (c) State 3, (d) State 4, (e) State 5, (f) State 6, and (g) State 7.

$$v_{C1}(t_2) = V_{in} - (V_{in} - v_{C1}(t_1)) \cdot \cos \omega_0(t_2 - t_1) + Z_0 \cdot i_{Ls}(t) \cdot \sin \omega_0(t_2 - t_1) \\ = v_{C1}(\text{PPK-IV}) \quad (161)$$

$$i_{ds1}(t) = i_{L1}(t) + i_{Ls}(t) \quad (162)$$

$$\Delta t_2 = (t_2 - t_1) = T_{on} - T_1 \quad (163)$$

3) State 3 (t_2-t_3): As displayed in Fig. 11(c), when Q_1 is turned off, the freewheeling currents i_{L1} and i_{L3} discharge C_1 . As a result, v_{ds1} gradually increases, reducing the cross-sectional area of i_{ds1} and v_{ds1} on Q_1 . The turn-off switching loss is reduced. The rise time of v_{ds1} is calculated as shown in (171). The freewheeling current of i_{Ls} is also demagnetized and D_5 is forward-biased. Terms i_{L1} , i_{L3} , i_{Ls} , i_{C1} , and v_{C1} are expressed in (164), (165), and (167). During state 3, v_{C1} reduces from V_o to 0. When v_{C1} is equal to zero, the converter transitions to State 4.

$$i_{L1}(t) = i_{L3}(t) = i_{L1}(t_2) + \left(\frac{V_{in}}{L_1 + L_3} \right) \cdot (t - t_2) = i_{L1}(t_2) + \frac{V_{in}}{L_1} \cdot (t - t_2) \quad (164)$$

$$i_{Ls}(t) = i_{Ls}(t_2) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t - t_2) \quad (165)$$

$$i_{C1}(t_3) = i_{C1}(t) = i_{L1-pk} \quad (166)$$

$$v_{C1}(t) = v_{C1}(t_2) + \left(\frac{-i_{L1}(t_2)(t - t_2)}{C_1} \right) \quad (167)$$

$$v_{C1}(t_3) = v_{C1}(t_2) + \left(\frac{-i_{L1}(t_2)(t_3 - t_2)}{C_1} \right) = 0 \quad (168)$$

$$v_{ds1}(t) = V_o - v_{C1}(t) \quad (169)$$

$$i_{D5}(t) = i_{L1}(t) + i_{Ls}(t) \quad (170)$$

$$\Delta t_3 = (t_3 - t_2) = \frac{C_1 \cdot v_{C1}(t_3)}{i_{L1-pk}} \quad (171)$$

4) State 4 (t_3-t_4): As displayed in Fig. 11(d), Q_1 is kept in the off-state and i_{L1} and i_{Ls} are continuously demagnetized. Because v_{C1} is zero, D_1 is forward-biased, causing i_{D1} to increase from 0. Terms i_{L1} , i_{Ls} , i_{C1} , and v_{C1} can be expressed in (172), (173), (175), and (176). During State 4, i_{L3}

and i_{C1} decrease. When i_{L3} and i_{C1} reach zero, the converter transitions to State 5.

$$i_{L1}(t) = i_{L1}(t_3) + \left(\frac{V_{in} - V_o}{L_1} \right) \cdot (t - t_3) \quad (172)$$

$$i_{Ls}(t) = i_{Ls}(t_3) + \left(\frac{V_{in} - V_o}{L_s} \right) \cdot (t - t_3) \quad (173)$$

$$i_{L3}(t) = -i_{C1}(t) = i_{L1-pk} \cdot \cos \omega_2(t - t_3) + \frac{v_{C1}(t_3)}{Z_2} \cdot \sin \omega_2(t - t_3) \quad (174)$$

$$i_{C1}(t_4) = -i_{L3}(t_4) = 0 \quad (175)$$

$$v_{C1}(t) - \sqrt{\frac{(-i_{L1-pk})^2 \cdot L_3}{C_1}} \cdot \sin \omega_2(t - t_3) = v_{C1}(t) \quad (176)$$

$$v_{C1}(t_4) = v_{C1}(t_3) - \sqrt{\frac{(-i_{L1-pk})^2 \cdot L_3}{C_1}} = v_{C1}(\text{NPK-IV}) \quad (177)$$

$$v_{ds1}(t) = V_o - v_{C1}(t) \quad (178)$$

$$v_{ds1}(t_4) = V_o - v_{C1}(t_4) = v_{ds1-pk} = V_o - v_{C1}(\text{NPK-IV}) \quad (179)$$

$$i_{D5}(t) = i_{L1}(t) - i_{C1}(t) \quad (180)$$

$$i_{D1}(t) = i_{L1}(t) + i_{C1}(t) \quad (181)$$

$$\Delta t_4 = (t_4 - t_3) = \frac{T_2}{4} \quad (182)$$

5) State 5 (t_4-t_5): As displayed in Fig. 11(e), Q_1 is kept in the off-state and i_{L1} and i_{Ls} are continuously demagnetized. In this state, L_3 , L_s , and C_1 resonate. Since L_3 is much smaller than L_s , the inductance of L_3 and its voltage v_{L3} can be ignored when the cascaded L_s and L_3 resonate with C_1 . Therefore, the resonant tank can be described simply as C_1 resonating with L_s , i_{C1} and v_{C1} are expressed in (184) and (185), respectively. When i_{C1} is discharged to 0, D_3 will be reverse biased. The converter then transitions to State 6.

$$i_{L1}(t) = i_{L1}(t_4) + \left(\frac{V_{in} - V_o}{L_1} \right) \cdot (t - t_4) \quad (183)$$

$$i_{Ls}(t) = i_{L3}(t) = i_{C1}(t) \\ = i_{Ls}(t_4) \cdot \cos \omega_0(t - t_4) + \frac{(V_{in} - V_o - v_{C1}(t_4))}{Z_0} \cdot \sin \omega_0(t - t_4) \quad (184)$$

$$v_{C1}(t) = (V_{in} - V_o) - ((V_{in} - V_o) - v_{C1}(t_4)) \cdot \cos \omega_0(t - t_4) \\ + Z_0 \cdot i_{Ls}(t_4) \cdot \sin \omega_0(t - t_4) \quad (185)$$

$$v_{ds1}(t) = V_o \quad (186)$$

$$\Delta t_5 = (t_5 - t_4) = \frac{C_1 \cdot (0 - v_{C1}(t_4))}{i_{Ls}(t_4)} \quad (187)$$

6) State 6 (t_5 – t_6): As displayed in Fig. 11(f), Q_1 is kept in the off-state. Since v_{C1} is charged to V_o , the freewheeling current of i_{Ls} flows through D_5 . i_{L1} and i_{Ls} can be expressed as (188) and (189), respectively. i_{L3} and i_{C1} are both 0. When i_{Ls} is demagnetized to zero, D_5 is reverse biased. Then, the converter transitions to State 7.

$$i_{L1}(t) = i_{L1}(t_5) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_5) \quad (188)$$

$$i_{Ls}(t) = i_{D5}(t) = i_{Ls}(t_5) + \left(\frac{V_{in} - V_o}{L_s} \right) (t - t_5) \quad (189)$$

$$\Delta t_6 = (t_6 - t_5) = \frac{L_s \cdot (0 - i_{Ls}(t_5))}{V_{in} - V_o} \quad (190)$$

7) State 7 (t_6 – $t_0 + T_{SW}$): As displayed in Fig. 11(g), Q_1 is kept in the off-state and D_1 is kept in the on-state. In this condition, there is no current flowing through any of the components of the snubber. Simultaneously, the energy stored in L_1 is transferred to the output. Hence, i_{L1} decreases gradually. The converter returns to State 1 when Q_1 is turned on.

$$i_{L1}(t) = i_{D1}(t) = i_{L1}(t_6) + \left(\frac{V_{in} - V_o}{L_1} \right) (t - t_6) \quad (191)$$

$$v_{C1}(t) = v_{C1}(t_6) = 0 \quad (192)$$

$$\Delta t_6 = (t_0 + T_{SW} - t_6) = T_{SW} - t_6 \quad (193)$$

3. Design of the prototype circuit

Fig. 12(a) displays a power supply unit (PSU) for DC-powered network equipment. The PSU consists of the front-end boost converter and the second-stage isolated DC-DC converter. The front-end boost can raise the low input voltage and store sufficient energy in C_o to achieve the required hold-up time.

Fig. 12(b) displays the control block diagram of the prototype, which is based on a commercial boost controller with typical peak current mode control. The feedback error amplifier, derived from an operational transconductance amplifier (OTA), functions as a voltage-controlled

Table 3

Circuit specifications.

Symbol	Definition	Value
V_{in}	Input voltage of boost stage	20 V~45 V DC
I_{in}	Input current of boost stage	0–15 A
V_o	Output voltage of boost stage	54 DC
P_o	Rated output power of boost stage	270 W
$I_{o-rated}$	Rated output current of boost stage	5 A
f_{sw}	Switching frequency of boost stage	95 kHz
T_{sw}	Switching period of boost stage	10.526 μ s
η_{bst}	Expected efficiency of boost stage	95 % at full load
η_{PSU}	Expected efficiency of PSU	90 % at full load

current source (VCCS). It also includes a built-in low-voltage dropout regulator, a programmable soft-start timer, slope compensation, and an overvoltage protection function. The utilization of a current transformer as a current sensor helps minimize power losses.

Table 3 lists the design specifications of the PSU, while Table 4 displays the specifications of the components used in the prototype verification circuit for the snubber. The Q_1 from ON Semiconductor Corp. is chosen for its cost-effectiveness. The D_1 is made by Monolithic Power Systems Inc.

The first when designing the prototype is to determine Δi_{L1} as 25 % of the output current, based on industrial design experience. The ideal value of L_1 is calculated as 37.7 μ H using (196) to determine the number of initial winding turns, which is 13.24. This value is rounded up to the nearest integer, which is 14. The inductance of L_1 is then recalculated to 42.1 μ H, as displayed in (198). The peak current and maximum flux density are calculated as (199) and (201), respectively. The maximum flux density of 0.384 T represented 93.7 % of the high-temperature saturation flux density of 0.41 T for the JPP95 material. This achieves the industrial design requirement with a 95 % margin. The core and copper losses of L_1 are calculated as (204) and (205), respectively.

Table 4

Specifications of key components.

Symbol	Components description	Manufacturer
Q_1	FDP085N10A, 100 V, $R_{ds1(on)}=8.5$ m Ω , $C_{oss1}=752$ pF	ON semi
D_o	MP9989A, SR-based ideal diode, 10 m Ω	MPS, Inc.
D_1	SS110, 100 V/1 A, $V_{F,D1}=0.8$ V	Taiwan Semi
D_2	SS310, 100 V/3 A, $V_{F,D2}=0.8$ V	Taiwan Semi
L_1	42.1 μ H, PQ2620 with air gap, Material JPP95, $B_s = 0.41$ T(at 100 °C), $A_e \cdot L_1 = 122.6$ mm ² , $V_{e-L1} = 5440$ mm ³ , $A_{L1} = 215$ nH/N ² , $R_{L1} = 5.6$ m Ω	A-CORE Co.
L_3	IHL P-2525CZ, 220 nH, $R_{L3} = 2.5$ m Ω	Vishay
L_s	47 μ H, SRN8040HA-470 M, $R_{Ls} = 0.19$ Ω	Bourns
C_{in}	470 μ F/63 V, E-cap	Rubycon
C_1	10 nF $\times$ 4, 100 V, MLCC, $R_{C1} = 20$ m Ω , $L_{C1} = 1.05$ nH	Murata
C_o	2 $\times$ 2200 μ F/63 V, $R_{Co}=8.5$ m Ω , E-cap	Rubycon
U_1	RT8525, peak current mode PWM IC	Richtek

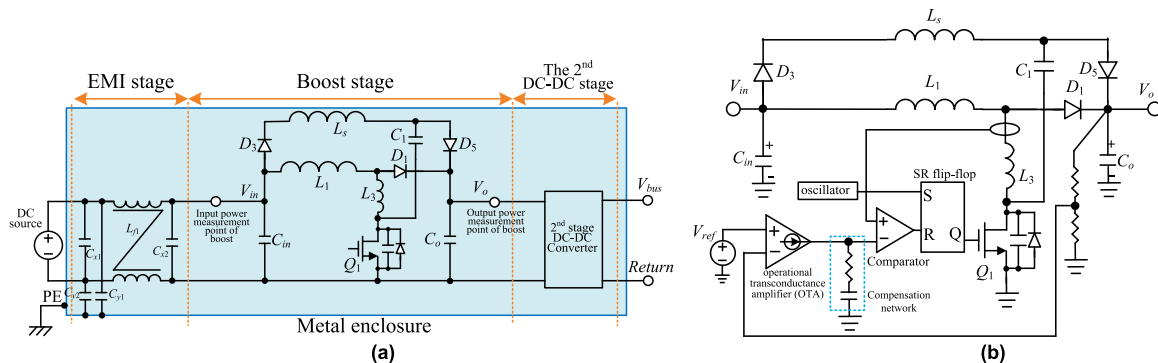


Fig. 12. System blocks: (a) Main blocks of the PSU; (b) Control block diagram of the boost stage.

$$D_{max} = 1 - \frac{V_{in(min)}}{V_o} = 0.63 \quad (194)$$

$$\Delta i_{L1} = 0.25 \cdot I_{o-rated} = 3.52A \quad (195)$$

$$L_1 = \frac{V_{in(min)} \cdot D_{max}}{f_{sw} \cdot \Delta i_{L1}} = 37.7 \mu H \quad (196)$$

$$N_{L1} = \sqrt{\frac{L_1}{A_{L1}}} = 13.24 \text{ turns} \Rightarrow 14 \text{ turns} \quad (197)$$

$$L_1 = \sqrt{A_{L1} \cdot (14)^2} = 42.1 \mu H \quad (198)$$

$$i_{L1-pk} = \frac{\Delta i_{L1}}{2} + \frac{P_{in}}{V_{in(min)}} = 15.64A \quad (199)$$

$$i_{L1-vly} = i_{L1-pk} - \Delta i_{L1} = 12.49A \quad (200)$$

$$B_{max} = \frac{L_1 \cdot i_{L1-pk}}{A_e \cdot N_{L1}} = 0.384T \quad (201)$$

$$CoreLoss_{(380mT-100kHz)} = 1000 \frac{mW}{cm^3} \quad (202)$$

$$i_{L1-rms} = \sqrt{\left(i_{L1-vly} + \frac{\Delta i_{L1}}{2}\right)^2 + \frac{\Delta i_{L1}^2}{12}} = 14.09A \quad (203)$$

$$P_{L1-core} = CoreLoss_{(380mT-100kHz)} \cdot V_{e-L1} = 5.44W \quad (204)$$

$$P_{L1-copper} = (i_{L1-rms})^2 \cdot R_{L1} = 1.11W \quad (205)$$

The parameters of the snubber are designed as follows. According to (206) with a desired rise time t_{r-vds1} of 150 ns, the capacitance of C_1 is determined as 40 nF. Considering the high pulse current on C_1 , as demonstrated in (199), C_1 is implemented using four parallel 10 nF MLCCs. To simplify the design process, a 47 μH SMD inductor is used for L_s . The resonant frequency and period are (207) and (208), respectively.

$$C_1 = i_{L1-pk} \cdot \frac{t_{r-vds1}}{V_o} - C_{oss1} = 42.7nF \quad (206)$$

$$\omega_0 = \frac{1}{\sqrt{C_1 \cdot L_s}} = 7.293 \times 10^5 \text{ rad/sec} \quad (207)$$

$$\frac{T_0}{2} = \pi \sqrt{C_1 \cdot L_s} = 4.31 \mu s \quad (208)$$

The positive voltage peak $v_{C1(PPk)}$ of v_{C1} in the four operation modes is determined from (209), (210), and (211). The negative voltage peak $v_{C1(NPk)}$ of v_{C1} in different operational modes is determined using (212). From (212), it is evident that the negative voltage of v_{C1} is determined by the selected values of L_3 and C_1 . C_1 is determined from (206). The next step is to determine the value of inductance for L_3 . According to industrial design requirements, MOSFETs have a de-rating of 95 %, which

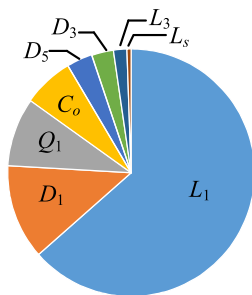


Fig. 13. Loss breakdown analysis of boost converter.

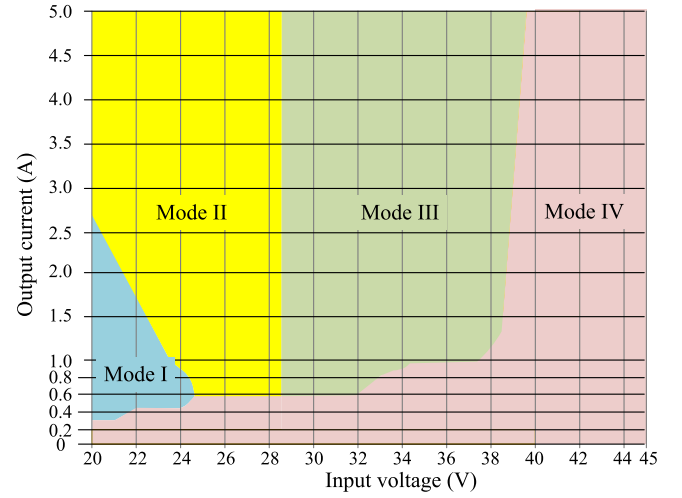


Fig. 14. Regions of operation modes.

limits their maximum operating voltage to 95 V from a potential 100 V. Eq. (212) demonstrates that the most extreme negative voltage peak occurs when the input voltage is at its lowest limit and the device is operating at full load. Assuming that L_3 is 220 nH and substituting it into (215), $v_{C1(NPk)}$ is equal to -36.7 V. This value results in a $v_{ds1(pk)}$ value of 90.7 V, which is still below the maximum operating voltage of 95 V. v_{ds1-pk} is 90.7 V in (216). Therefore, the design requirement (Application note) of de-rating MOSFETs rated at 100 V by 95 % is achieved.

$$v_{C1(PPk-I)} = 2 \cdot V_{in} - \left(-\sqrt{\frac{L_3 \cdot (i_{L1-pk})^2}{C_1 + C_{oss1}}} \right) \leq V_o \quad (209)$$

$$v_{C1(PPk-II)} = v_{C1(PPk-III)} = V_o \quad (210)$$

$$v_{C1(PPk-IV)} \approx V_o - V_{in} \cdot \cos(\omega_0 \cdot T_{on}) < V_o \quad (211)$$

$$v_{C1(NPk-I)} = v_{C1(NPk-II)} = v_{C1(NPk-III)} = v_{C1(NPk-IV)} = -\sqrt{\frac{(i_{L1-pk})^2 \cdot L_3}{C_1 + C_{oss1}}} = v_{C1(NPk)} \quad (212)$$

$$v_{ds1-pk} = V_o - v_{C1(NPk)} \leq 95V \quad (213)$$

$$v_{C1(NPk)} \geq -41V \quad (214)$$

$$v_{C1(NPk)} = -\sqrt{\frac{(i_{L1-pk})^2 \cdot L_3}{C_1 + C_{oss1}}} = -36.33V \quad (215)$$

$$v_{ds1-pk} = V_o - v_{C1(NPk)} = 90.33V \quad (216)$$

The desired rise time t_{r-vds1} is 150 ns, which is significantly longer than the 17.2 ns calculated by (217). Therefore, the switching loss of Q_1 could be reduced when it is turned off. By analyzing (218) and (219), it is evident that the rise time of i_{ds1} in Q_1 is 50.9 ns. This value is equivalent to 30.6 % of the turn-on time of Q_1 , indicating that the switching loss could also be reduced when Q_1 is turned on. The total loss of Q_1 is calculated using (224). The power losses of L_3 , C_0 , L_s , D_3 , D_5 , and D_1 are listed as (227), (229), (232), (234), (236), and (237), respectively. The total estimated power loss is 10.318 W from (238).

$$t_{Q1-turn-off} = \frac{\left(Q_{gd1} + \frac{Q_{gs1}}{2}\right) (R_{g1(int)} + R_{g1(off-ext)})}{V_{th1}} = 17.2ns \quad (217)$$

$$t_{L3-turn-on} = \frac{L_3 \cdot (0 - i_{L1-vly})}{0 - V_o} = 50.9ns \quad (218)$$

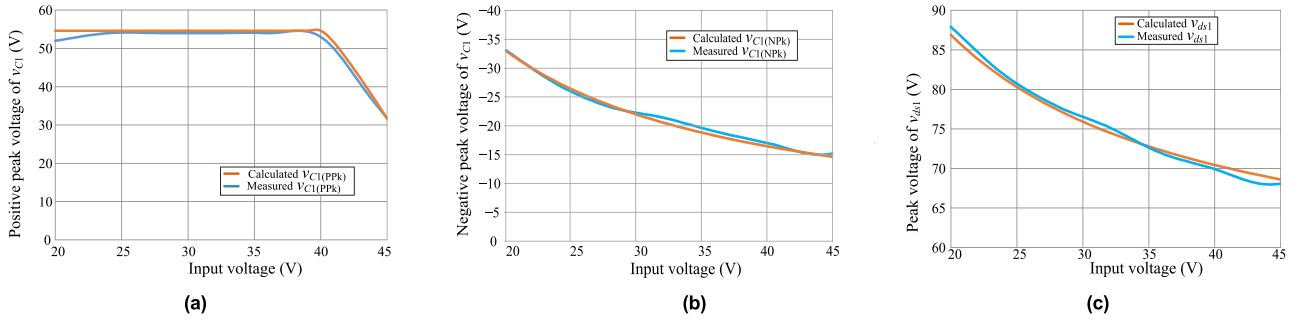


Fig. 15. Calculation and experimental results: (a) Positive peak voltage of v_{C1} , (b) Negative peak voltage of v_{C1} , and (c) v_{ds1-pk} .

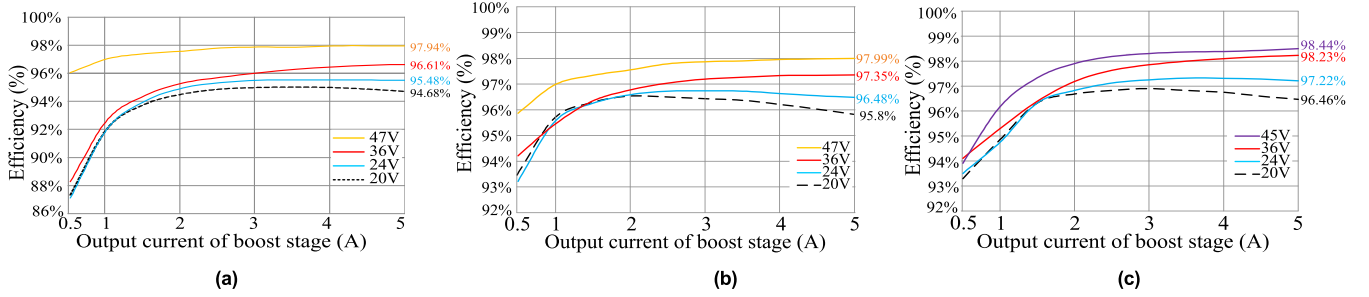


Fig. 16. Experimental results of conversion efficiency: (a) the original boost converter; (b) the snubber circuit in (Yau and Hung, 2023), and (c) the proposed snubber.

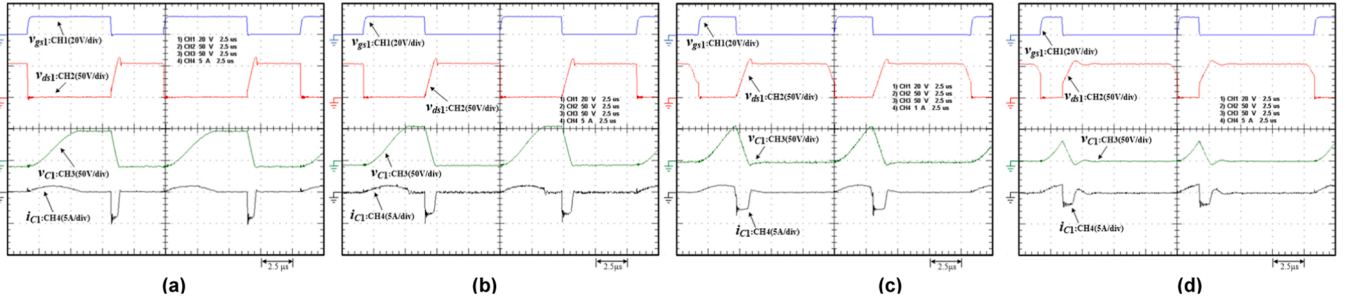


Fig. 17. Waveforms of v_{gs1} , v_{ds1} , v_{C1} , and i_{C1} of the proposed boost converter of 20 % load: (a) $V_{in} = 20$ V, (b) $V_{in} = 28$ V, (c) $V_{in} = 36$ V, and (d) $V_{in} = 45$ V.

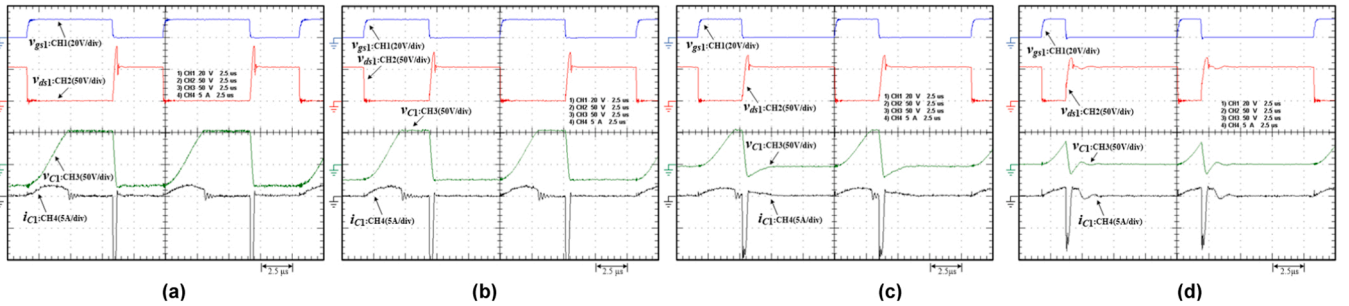


Fig. 18. Waveforms of v_{gs1} , v_{ds1} , v_{C1} , and i_{C1} of the proposed boost converter at 100 % load: (a) $V_{in} = 20$ V, (b) $V_{in} = 28$ V, (c) $V_{in} = 36$ V, and (d) $V_{in} = 45$ V.

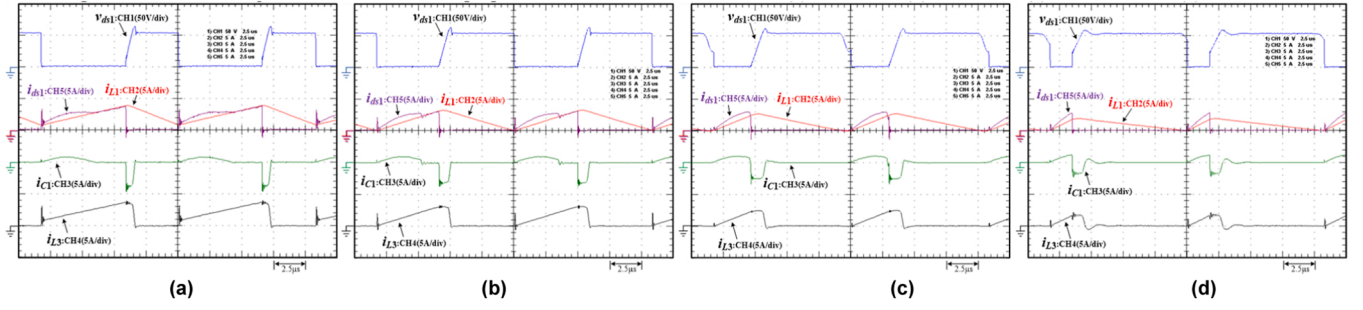


Fig. 19. Waveforms of v_{ds1} , i_{L1} , i_{C1} , i_{L3} , and i_{ds1} of the proposed boost converter at 20 % load: (a) $V_{in} = 20$ V, (b) $V_{in} = 28$ V, (c) $V_{in} = 36$ V, and (d) $V_{in} = 45$ V.

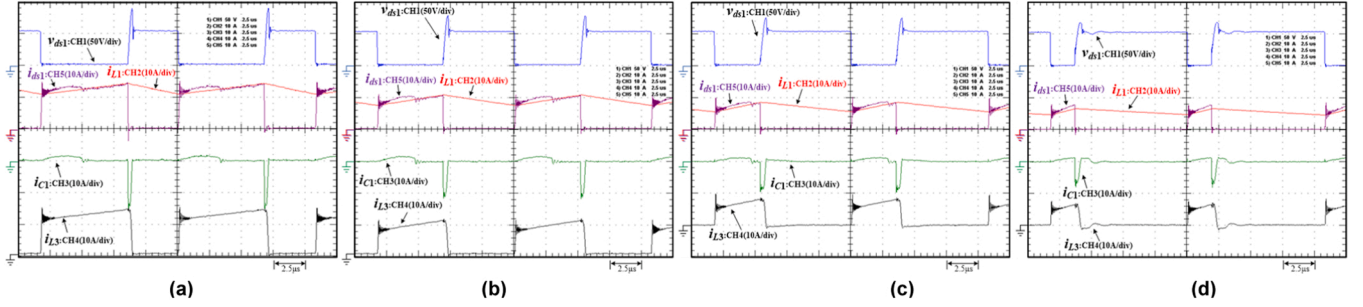


Fig. 20. Waveforms of v_{ds1} , i_{L1} , i_{C1} , i_{L3} , and i_{ds1} of the proposed boost converter under 100 % load: (a) $V_{in} = 20$ V, (b) $V_{in} = 28$ V, (c) $V_{in} = 36$ V, and (d) $V_{in} = 45$ V.

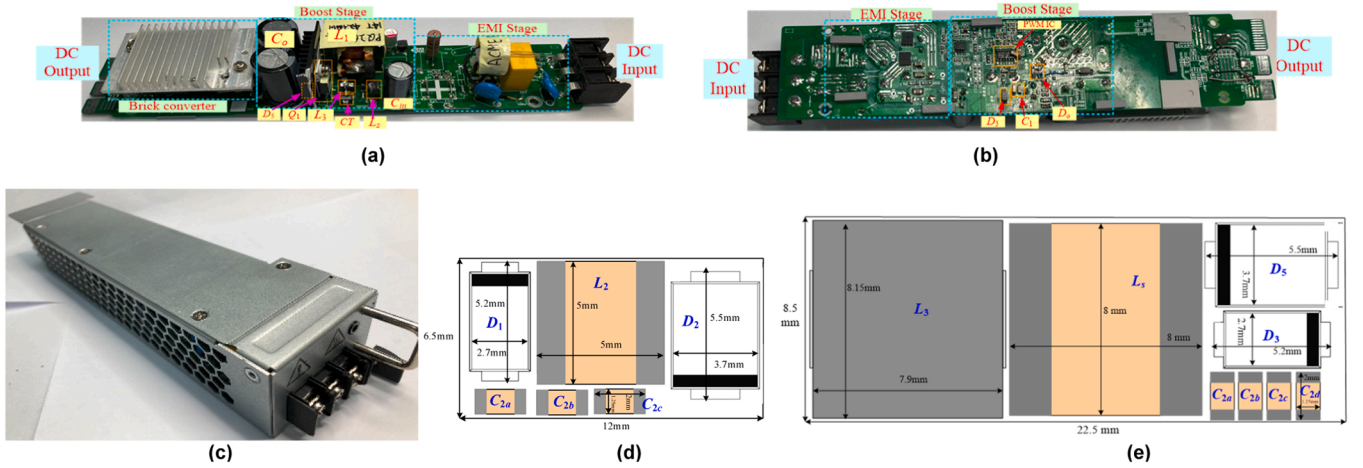


Fig. 21. Prototype photographs of the proposed boost converter: (a) top view of PSU, (b) bottom view of PSU, (c) complete PSU module, and (d) area of the snubber in (Yau and Hung, 2023), and (e) area of the proposed snubber.

$$t_{Q1-turn-on} = \frac{\left(Q_{gd1} + \frac{Q_{g1}}{2}\right)(R_{g1(int)} + R_{g1(on-ext)})}{V_{DR1} - V_{th1}} = 15.6\text{ns} \quad (219)$$

$$P_{Q1-sw} = \frac{1}{2} \frac{(t_{Q1-turn-on})^2}{t_{L3-turn-on}} V_o f_{SW} i_{L1-vly} + \frac{1}{2} \frac{(t_{Q1-turn-off})^2}{t_{r-vds1}} V_o f_{SW} i_{L1-pk} + \frac{C_{oss1} \cdot V_o^2 \cdot f_{SW}}{2} = 0.228\text{W} \quad (220)$$

$$P_{Q1-drv} = Q_{g1} \cdot V_{DR1} f_{SW} = 30.4\text{mW} \quad (221)$$

$$i_{ds1-rms} = \sqrt{D_{max} \left(i_{L1-vly} + \frac{\Delta i_{L1}}{2} \right)^2 + \frac{\Delta i_{L1}^2}{12}} = 8.9\text{A} \quad (222)$$

$$P_{Q1-cond} = (i_{ds1-rms})^2 \cdot R_{ds1(on)} = 0.673\text{W} \quad (223)$$

$$P_{Q1-tot} = P_{Q1-cond} + P_{Q1-sw} + P_{Q1-drv} = 0.932\text{W} \quad (224)$$

$$i_{L3-pk} = i_{L1-pk} = 15.64\text{A} \quad (225)$$

$$i_{L3-rms} = \sqrt{(1 - D_{max}) \left(i_{L1-vly} + \frac{\Delta i_{L1}}{2} \right)^2 + \frac{\Delta i_{L1}^2}{12}} = i_{D1-rms} = 8.58\text{A} \quad (226)$$

$$P_{L3} = (i_{L3-rms})^2 \cdot R_{L3} = 0.184\text{W} \quad (227)$$

$$i_{Co-rms} = \sqrt{(i_{D1-rms})^2 - (I_{o-rated})^2} = 6.97\text{A} \quad (228)$$

$$P_{Co} = (i_{Co-rms})^2 \cdot R_{Co} = 0.68\text{W} \quad (229)$$

$$i_{Ls-pk} = \sqrt{\frac{C_1 [(v_{C1(PPk-II)})^2 + (v_{C1(NPk)})^2]}{L_s}} = 1.9A \quad (230)$$

$$i_{Ls-rms} = \frac{i_{Ls-pk}}{\sqrt{2}} \cdot \left(\frac{1}{2} \frac{T_0}{T_{sw}} \right) = 0.551A \quad (231)$$

$$P_{Ls} = (i_{Ls-rms})^2 \cdot R_{Ls} = 58mW \quad (232)$$

$$i_{D3-avg} = \frac{(v_{C1(PPk-II)} - v_{C1(NPk)}) \cdot C_1}{T_{SW}} = 0.345A \quad (233)$$

$$P_{D3} = i_{D3-avg} \cdot V_{F-D3} = 0.293W \quad (234)$$

$$i_{D5-avg} = \frac{(v_{C1(PPk-II)} - v_{C1(NPk)}) \cdot C_1}{T_{SW}} = 0.345A \quad (235)$$

$$P_{D5} = i_{D5-avg} \cdot V_{F-D5} = 0.345W \quad (236)$$

$$P_{D1} = R_{D1} (i_{D1-rms})^2 + f_{sw} \cdot Q_{rr-D1} \cdot V_o = 1.28W \quad (237)$$

$$P_{tot} = P_{L1-core} + P_{L1-copper} + P_{Q1-tot} + P_{Ls} + P_{L3} + P_{D1} + P_{D3} + P_{D5} + P_{Co} \\ = 10.318W \quad (238)$$

The breakdown of loss analysis of the boost converter are shown in Fig. 13. L_1 and Q_1 contributed 63.5 % and 9 % of the total loss, respectively, while D_1 and C_o contribute 12.4 % and 6.6 % of the total loss, respectively. The loss of the remaining components is less than 3 %.

The map of regional operating modes obtained through simulation for various input voltages and output powers is displayed in Fig. 14. Here, it can be observed that Mode I only occurs at a low input voltage with a light load, while Mode IV occurs during a short T_{on} time. Therefore, Mode IV could occur under the conditions of low input voltage with a light load, high input voltage with a light load, and high input voltage with a heavy load operation. It should be noted that there is no direct boundary between Mode I and Mode IV. The transition between Mode I and Mode IV must be made via Mode II and Mode III, as displayed in the state machine diagram in Fig. 3.

4. Experimental results of the prototype circuit

The curves of the positive peak voltage of v_{C1} in Fig. 15(a) are obtained from (28), (70), (108), (161), and the measurement results. Based on (212) and the actual measurement results, the curve of the negative peak voltage of v_{C1} is plotted in Fig. 15(b). From the experimental results, it is revealed that the calculated values closely matched the measured values. In Fig. 15(c), the theoretically calculated value of v_{ds1-pk} and the actual measured value are revealed as quite consistent.

Figs. 16(a) and (b) display the measured results of the conversion efficiency of the original boost converter and the reference circuit (Yau and Hung, 2023), respectively. The conversion efficiency of the prototype circuit is displayed in Fig. 16(c). Compare to reference (Yau and Hung, 2023), it can be found that when V_{in} is 20 V at rated load, the proposed method can increase the conversion efficiency by 0.65 %. When V_{in} is 36 V at rated load, the proposed method can increase the conversion efficiency by 0.92 %.

Fig. 17 displays the waveforms of v_{gs1} , v_{ds1} , v_{C1} , and i_{C1} for the prototype circuit under varying input voltages and a 20 % load. Fig. 18 displays the waveforms of v_{gs1} , v_{ds1} , v_{C1} , and i_{C1} under varying input voltages and full load. Figs. 17(a), (b), (c), and (d) correspond to Modes I, II, III, and IV, respectively. These modes correspond to the simulation results displayed in Fig. 14.

Figs. 18(a), (b), (c), and (d) refer to Modes II, II, III, and IV, respectively. These modes correspond to the simulation results displayed in Fig. 14. The waveforms presented in Fig. 18(a) are also in close

agreement with the design results obtained from (194)–(231).

Fig. 19 displays the waveforms of v_{ds1} , i_{L1} , i_{C1} , i_{L3} , and i_{ds1} for the prototype circuit under varying input voltages and 20 % load. Fig. 20 displays the waveforms of v_{ds1} , i_{L1} , i_{C1} , i_{L3} , and i_{ds1} under varying input voltages and full load. Figs. 19(a), (b), (c), and (d) correspond to Modes I, II, III, and IV, respectively. These measured waveforms are consistent with the analysis presented in Figs. 4, 6, 8, and 10, respectively.

Figs. 20(a), (b), (c), and (d) refer to Modes II, II, III, and IV, respectively. These modes correspond to the simulation results displayed in Figs. 4, 6, 8, and 10, respectively. The waveforms in Fig. 20(a) are also in close agreement with the design results obtained from (194)–(231).

Figs. 21(a) and (b) display the top and bottom views of the prototype circuit, respectively. Fig. 21(c) displays the complete PSU, which can be used for networking equipment, such as a power supply in the disaggregated cell site gateway (DCSG) router (Datasheet). The components referenced in (Yau and Hung, 2023) require a small PCB space with a total area of 78 mm², as depicted in Fig. 21(d). However, the proposed snubber requires only an additional PCB space of 113.5 mm², as shown in Fig. 21(e), but offers significant efficiency improvement.

5. Conclusion

In this article, a new soft-switching snubber for a boost converter is proposed. Compared to previous studies, the proposed method relies entirely on passive components, which is a significant advantage.

The proposed snubber is suitable for converter topologies in which the input is a current source, such as boost DC-DC converters or boost-based power factor correction converters. The proposed snubber can achieve soft switching at both the on and off edges of the main switch. This novel approach increases the conversion efficiency from 0.65 % to 0.92 % with only an 113.5 mm² increase in the component area compared to the reference configuration (Yau and Hung, 2023).

In this article, the proposed circuit is analyzed and theoretically derived. Subsequently, a prototype is designed, simulated, and validated based on the preceding theoretical analysis. The proposed circuit can be operated by a traditional analog PWM control IC without the need for any digital control or additional switches, reducing system complexity and enhancing the cost-effectiveness of commercial products.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yeu-Torng Yau reports financial support was provided by National Science and Technology Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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